

The Geometry of the Accusative

Abstract

This paper challenges the existence of structural Case as an independent syntactic primitive within the Minimalist Program, arguing against both the Case-by-Agree mechanism and Dependent Case Theory. Adopting a strictly spatial-relational framework, we argue that unmarked core arguments (Nominative and Absolutive) are inherently caseless roots that saturate the eventive core through topological fusion, or pseudo-incorporation. Consequently, the Case Filter and Burzio's Generalization are revealed to be theoretical artifacts. We propose that marked alignments, such as Differential Object Marking (DOM) and the Ergative case, are not the result of abstract feature valuation or *phase-mate* competition, but the overt materialization of elementary spatial boundaries. Highly individuated internal arguments are shielded from eventive fusion via the Zonal Inclusion operator ($\subseteq$), mapping them as topological targets or possessors, while active external arguments are anchored to the event perimeter via the Origin/Reverse Inclusion vector ($\ni$). By replacing abstract functional heads (rendering *little-v* obsolete) with directly interpretable topological relators, this architecture captures cross-linguistic syncretisms, aspectual splits, and global case splits without resorting to invisible probes or the stipulative geometric hierarchies of Nanosyntax.

Keywords: Accusative, Differential Object Marking, Dependent Case Theory, Pseudo-incorporation, Zonal inclusion.

1. Introduction: Background

The foundational intuition of the Minimalist Program (Chomsky 1995) was predicated on a radical simplification of the language architecture. Under the Strong Minimalist Thesis, the computational system of human language operates as an optimal solution to legibility conditions imposed by the Sensorimotor (S-M) and Conceptual-Intentional (C-I) interfaces. In its purest conceptualization, syntax operates strictly through the recursive application of a single, costless operation *Merge* combining substantive lexical items to build interpretable hierarchical structures without superfluous derivational steps or *ad-hoc* representational filters.

The subsequent evolution of the framework, specifically regarding Case and Agreement, introduced a set of theoretical tensions. In order to account for the morphological distribution of nominal arguments, Chomsky (2000, 2001) assumes that nominals enter the derivation bearing an uninterpretable Case feature, [uCase]. Because uninterpretable features are illegible at the interfaces, they act as "derivational time-bombs" (Preminger 2014) that must be valued and deleted before Spell-Out. This deletion is achieved via the operation *Agree*, where a functional probe (such as *v* or T) carrying uninterpretable ϕ -features searches its c-command domain for a matching nominal goal.

While empirically wide-ranging, the *Case-by-Agree* mechanism undermines the original minimalist promise. It forces the syntax to project features whose sole *ontological* purpose is to be deleted. This architectural redundancy is particularly evident in the treatment of Differential Object

Marking (DOM). When confronted with the fact that many languages morphologically mark only a specific subset of direct objects (e.g., animate, definite, or specific), contemporary approaches resort to a massive proliferation of abstract functional heads and uninterpretable features to rescue the structural calculus.

For instance, in formalizing the differential marking of animate objects, scholars operating within the minimalist framework frequently postulate the existence of abstract [PERSON] or [δ] (discourse-linking) features that require specialized licensing above the standard vP. In systems like those proposed by Kalin (2018) or Irimia and Pineda (2021), a differentially marked object cannot simply be licensed by v; rather, it must raise to the specifier of an invisible intermediate projection, often labeled α P or ApplP, specifically generated to value its extra [PERSON] feature and its [uCase]. The geometric complexity required to derive a DOM object in these models is schematized in (1), adapted from López (2012) and Bárány (2018).

- (1) [vP vACC [α P DP [uCase, PERSON] [α' α [VP V DP_t]]]]

We think that this expansion leads to a set of conceptual contradictions when applied to cross-linguistic data. Consider the canonical DOM alternation in Hindi-Urdu, where an animate/specific object receives the overt accusative/dative marker =*ko*, while a non-specific object remains bare.

- (2) Hindi-Urdu (DOM alternation based on specificity)

- | | | | | |
|----|--|------------------|--------------|-------------|
| a. | Nadya=ne | jiraf | dekh-na | he |
| | Nadya.F.SG=ERG | giraffe.M.SG.NOM | see-INF.M.SG | be.PRS.3.SG |
| | ‘Nadya wants to see a giraffe/giraffes.’ | | | |
| b. | Nadya=ne | jiraf=ko | dekh-na | he |
| | Nadya.F.SG=ERG | giraffe.M.SG=ACC | see-INF.M.SG | be.PRS.3.SG |
| | ‘Nadya wants to see the giraffe.’ (adapted from Butt 1993: 90) | | | |

To derive the presence of =*ko* in (2b) using the standard Agree machinery, one must assume that the object *jiraf* carries an uninterpretable [uCase] feature, alongside an additional licensing need for its *specificity* that triggers movement to the α P projection. Conversely, the unmarked object in (2a) is either analyzed as undergoing pseudo-incorporation (lacking a Case feature entirely), or as receiving a "null" Case.

Alternatively, post-syntactic approaches in Distributed Morphology (Halle & Marantz 1993, Keine & Müller 2008) argue that *all* objects receive accusative Case in the syntax, but an

Impoverishment rule (arbitrarily) deletes the Case features on inanimate/non-specific objects at PF, blocking the insertion of the DOM marker. Similarly, Dependent Case Theory (DCT) (Marantz 1991; Baker 2015), often taken to be an elegant configurational alternative to Agree, fails when subjected to broader typological scrutiny. For instance, when presented with aspectual alignment splits where absolutely identical structural c-command relations between a subject and an object trigger the assignment of an Ergative case in the past perfective aspect but fail entirely in the present imperfective (cf. Manzini et al. 2015 on Punjabi), DCT algorithms falter. This exposes configurational calculus as a theoretical artifact rather than a reflection of human cognitive reality.

Furthermore, contemporary attempts to preserve the primacy of structural hierarchies have led to highly stipulative structural models. The highly sophisticated framework of Nanosyntax (Caha 2009), while deeply committed to representing morphological reality and cross-linguistic variation, rests its universal functional sequence (f-seq) upon an uninterpretable, allegedly foundational structural baseline: the existence of the Nominative or Absolutive case as a bundle of abstract features. However, overwhelming cross-linguistic evidence demonstrates unequivocally that unmarked core arguments are not complex feature bundles that have been assigned a "null" or default case; they are literally caseless, bare nominal roots. If the very base of the sequential structural architecture is a syntactic *phantom*, the entirety of the Nanosyntax project collapses into what can only be described as unnecessary functional stratification, artificially multiplying functional projections within the hierarchy itself to absorb empirical contradictions. Similar problems are increasingly evident in the biclausal hypothesis of split ergativity (Coon and Preminger 2017), which assume invisible syntactic boundaries and unpronounced auxiliary verbs to avoid conceding the failure of configurational case assignment.

We argue that neither of these solutions is truly minimalist. If, as we argue in the following sections, a DOM marker is an inherently meaningful spatial-relational predicate that merges directly with the nominal root to locate it as the *endpoint* (Goal/Possessor) of the event, the need for abstract probes, [uCase] valuation, and invisible α heads vanishes entirely. The object in (2b) is marked not because an abstract feature demanded deletion, but because the syntax deployed a relational/spatial vector to *tether* a highly salient entity to the transitive event.

Thus, we argue that a shared foundational error is the insistence on treating Case as an uninterpretable structural feature assigned via abstract functional heads. In this paper, we propose a radical departure from these *algebraic* viewpoints. We assume, backed by robust typological data, that "Structural Case" simply does not exist. Adopting a strictly spatial-relational framework founded firmly on the WYSIWYG (*What You See Is What You Get*) empirical principle, we argue

that the syntax of clausal transitivity is a continuous geometric reality governed exclusively by topological boundaries.

When the Conceptual-Intentional interface demands that an argument remain highly individuated, because it functions as a deictic center, a human entity, or a specific referent, the grammar actively prevents its topological fusion (pseudo-incorporation) with the eventive core by erecting overt *material* boundaries. It deploys the elementary relator of Zonal Inclusion ($\sqsubseteq$), (Belvin and den Dikken 1997; cf. Manzini and Franco 2016) to explicitly map the salient argument as the spatial terminus or "Possessor" of the event, thereby generating DOM. Conversely, the syntax deploys the spatial vector of Origin or Reverse Inclusion ($\sqsupseteq$) to firmly anchor a peripheral external participant as the mereological source of the eventive trajectory, surfacing cross-linguistically as the Ergative case (or a 'demoted' Agent).

By analyzing the "pure" Accusative as a caseless bare root and discarding the postulation of "action at a distance," this article aims to show that *all* morphologically realized cases are, without exception, self-licensing spatial/relational predicates. Transitivity is not an abstract algebra of invisible phi-probes assigned by invisible heads, but the geometric reality of human spatial/relational cognition applied to the architecture of events.

1.1 The Agree Fallacy

The theoretical tension within Minimalist accounts of DOM reaches its breaking point in the attempt to integrate differential marking into the *Case-by-Agree* calculus. Standard frameworks face an architectural puzzle when only a specific subset of objects (e.g., animate, definite) receives morphological marking. To resolve this without abandoning the configurational Case assignment module, recent proposals have engineered a system of "secondary licensing" mediated by abstract functional projections. As discussed above, one prominent strategy is to postulate invisible intermediate heads (e.g., αP or ApplP) specifically generated to establish an *Agree* relation at a distance with an abstract [PERSON] feature on the object (López 2012; Kalin 2018; Irimia & Pineda 2021).

Aware of the conceptual redundancy of postulating uninterpretable features and phantom projections strictly to trigger morphological reflexes, an alternative and highly sophisticated formalization has been advanced by Bárány (2017, 2018), building on the mechanism of *Cyclic Agree* (Béjar & Rezac 2009). In this ingenious framework, features are represented as complex, nested geometries (e.g., subset relations where 1st person contains 2nd, which contains 3rd). Under Cyclic Agree, a probe on v initiates its search downward. If the direct object is highly specified (e.g., a 1st/2nd person or animate nominal), it fully values and satisfies the probe, successfully

checking its case and halting the search. However, if the object is underspecified (e.g., a 3rd person or inanimate nominal), the probe on v is not entirely satisfied. As a result, the probe undergoes a "Cyclic Expansion", looking upward to agree with the external argument. Within this model, the presence of DOM is treated as a direct morphological reflex of the object's ability to fully satisfy the v probe in the first cycle, seemingly bypassing the need for additional αP heads.

However, despite its elegance, we argue the *Cyclic Agree* approach of Bárány and the postulation of nested feature geometries actually function as a complex formalization that obscures the underlying structural limitations of the model, its inherent inability to syntactically derive *referentiality*. The empirical vulnerability of this “bouncing probe” mechanism becomes evident when it is confronted with cross-linguistic data — data of which Bárány is clearly aware — that defy rigid, pre-compiled feature checking. Consider first the existence of fluid, pragmatically driven DOM systems, prominently attested in the Uralic family. In languages such as Eastern Khanty (Filchenko 2007), differential marking on the object is not statically triggered by an inherent [PERSON] or [ANIMACY] feature. Instead, it fluctuates purely based on discourse accessibility and the relative informational salience of the participants.

(3) Eastern Khanty (Pragmatically driven DOM)

- a. ma tām xo wěł-s-əm
 1SG this man kill-PST-1SG
 ‘I killed a/this man.’ (Object is New information, bare nominal)
- b. ma tām xo-ji wěł-s-əm
 1SG this man-DOM kill-PST-1SG
 ‘I killed the/this man.’ (Object is Given information, DOM) (Filchenko 2007: 399)

In order for a syntactic probe to compute fluid discourse dynamics phase-by-phase, the theory is forced to stipulate pragmatic notions like [TOPIC] or [GIVENNESS] are directly encoded as syntactic features within the nominal geometry that the v probe must satisfy. This constitutes a profound syntacticization of the discourse that violates the autonomy of syntax.

Even more problematic for the *Cyclic Agree* calculus is the existence of relative hierarchical DOM. In languages such as Awtuw (Papuan), the object is marked *only* if it outranks the subject on the animacy scale (Feldman 1986).

(4) Awtuw (Relative hierarchical DOM: 1/2 > 3Hum > Anim > Inan)

- a. Tey yaw d-ay-m

- 3FS pig FAC-bite-PST
 ‘She bit the pig.’ (Subj > Obj: No DOM)
- b. Yaw tey-e d-ay-m
 pig 3FS-DOM FAC-bite-PST
 ‘The pig bit her.’ (Obj > Subj: Obligatory DOM)
 (Feldman 1986: 110, see also Bárány 2018: 127)

To derive the Awtuw pattern via *Cyclic Agree*, it seems that the *v* probe would have to look downward at the object, delay its valuation, look upward at the subject, compare the nested feature geometries of both arguments, and then retroactively decide whether to assign DOM to the object. This implies a *Look-Ahead* violation: the syntax must "know" the feature composition of the external argument before concluding the licensing of the internal argument, overriding strict cyclicity.

These empirical problems dissolve if we abandon the idea of "action at a distance" and the arbitrary stacking of features. There is no need to assume probes that bounce up and down the clausal tree to compare animacy hierarchies. The computational system of human language resolves referentiality through a spatial topology. When the Conceptual-Intentional interface requires an argument to be highly salient, whether because it is a topic (Khanty), because it outranks the subject (Awtuw), or simply because it is an animate deictic center, the syntax refuses to pseudo-incorporate it as a bare, *inert* nominal root. Instead, it uses the geometry of space to establish a boundary, inserting a Zonal Inclusion operator, that we can label ($\subseteq$), following Manzini and Franco (2016) to *extract* the root and map it as the Goal or Possessor of the event. DOM is not the checking of a [PERSON] feature by an expanded probe; it is the fundamental spatial bounding of a salient entity.

1.2 The Dependent Case Alternative

A prominent minimalist alternative has been increasingly adopted to account for Differential Object Marking: Dependent Case Theory (DCT) (Marantz 1991; Baker & Vinokurova 2010; Preminger 2014; Baker 2015). Within DCT, case is not assigned by functional probes but is calculated configurationally at Spell-Out.

Under the standard DCT approach to DOM, as formalized by Baker and Vinokurova (2010) for Sakha, the structural calculus is tied to phase domains. A non-specific object remains within the lower *v*P/VP phase and receives no case. Conversely, a specific or animate object undergoes object shift, moving out of the VP and into the higher phase. Once in the same local domain as the subject,

the object is asymmetrically c-commanded by it, triggering the assignment of dependent Accusative case.

While Dependent Case Theory (DCT) elegantly eliminates the need for invisible [PERSON] probes, it reduces DOM to a structural reflex of c-command, generating empirical paradoxes. We argue that the fundamental flaw of the DCT algorithm is its absolute reliance on *phase-mate* competition and mandatory Object Shift. First, if DOM were strictly the dependent accusative assigned when an object vacates the VP to share a phase with a c-commanding subject, DOM should be structurally impossible in clauses lacking an external argument. Yet, as robustly attested in Romance impersonal and middle-passive constructions (e.g., Spanish *Se entiende a mi hijo* 'My son is understood'), DOM remains obligatory despite the complete absence of a competing subject DP in the local phase (cf. Manzini and Franco 2016). If case valuation is merely a counting calculus between two nominals, the emergence of DOM in single-argument domains remains theoretically inexplicable under the DCT framework.

Second, the mechanical requirement that marked objects must structurally shift to the vP edge to receive Case is empirically falsified by coordination asymmetries. As Kalin and Weisser (2019) show, numerous DOM languages freely allow the asymmetric coordination of unmarked and marked objects within the same complement position (e.g., Spanish *Vi [un asesino] y [a tres abogados]* 'I saw a murderer and DOM three lawyers'). If DOM valuation strictly required the marked object to extract from the VP phase, such constructions would induce massive violations of the Coordinate Structure Constraint, rendering the sentence ungrammatical.

These issues reveal that Differential Object Marking is not a dependent structural flag triggered by phase configuration. Instead, it is a topological boundary,¹ the vector of Zonal Inclusion ($\sqsubseteq$), deployed *in situ* to shield a salient Deictic Center from eventive fusion, operating entirely independently of any abstract phase-counting algorithm.

Furthermore, by analyzing DOM merely as a dependent structural Accusative, DCT is entirely blind to the spatial and semantic nature of the marker itself. The pervasive cross-linguistic syncretism between DOM and Dative/Allative morphology (Manzini & Franco 2016), prominently attested across Romance, Indo-Aryan, and Uralic, must be dismissed by DCT proponents as

¹ Throughout this work, we deliberately employ the term 'topological' rather than merely 'spatial' or 'structural' to define our relational architecture. By adopting a topological perspective, we argue that the syntactic spine is not a mechanical sequence of empty functional heads, but a geometric reality governed by boundaries, continuity, and inclusion. Topology abstracts away from literal three-dimensional physical containment. As Belvin and den Dikken (1997: 170) note regarding the concept of 'zonal inclusion', an object or eventuality may be included in a 'zone' associated with an entity without being physically contained within it. This topological abstraction allows us to unify physical location, abstract possession, and eventive transitivity into a single geometric primitive. Furthermore, it naturally formalizes the mechanics of argument licensing: non-salient, a-categorical roots undergo direct topological fusion (pseudo-incorporation) with the eventive core, whereas highly individuated arguments compel the syntax to project an explicit topological boundary (the $\sqsubseteq$ / $\sqsupseteq$ relators) to prevent this absorption and anchor them to the event.

massive, accidental homophony. Even more problematically, oblique DOM frequently induces syntactic restrictions, such as Person-Case Constraint (PCC) effects (Irimia 2021), that strictly differentiate it from purely structural accusatives. DCT's insistence on stripping DOM of its inherent relational content fails to capture its true syntactic profile.

Thus, we will argue that we can abandon the notion of abstract structural case entirely. Transitivity is not a local domain for calculating dependent features, but a spatial axis, as illustrated in (5).

(5) The Transitive Axis

[Origin/Source ($\supseteq$)] $\rightarrow$ EVENT $\rightarrow$ [Inclusion/Goal ($\subseteq$)]

We assume that the "pure" (unmarked) accusative object does not possess Case, nor does it wait for dependent calculus. It is a bare, a-categorial nominal root ($\emptyset$) that saturates the eventive space by topological fusion (pseudo-incorporation).

Differential Object Marking emerges as a fundamental necessity of spatial bounding. When the Conceptual-Intentional interface demands that an internal argument remain highly individuated (because it is a deictic center, a human, or a discourse-prominent entity) the syntax refuses to absorb it into the eventive core. To prevent this fusion, the grammar extracts the root and provides it with a distinct spatial coordinate by merging the object with the Zonal Inclusion ($\subseteq$) operator. By applying the ($\subseteq$) relator, the syntax explicitly maps the salient argument as the topological terminus, the Goal, or the "Possessor" of the event. This model effortlessly explains why DOM morphology is cross-linguistically syncretic with the Dative and the Allative: they are all manifestations of the exact same elementary predicate of Inclusion ($\subseteq$).

Symmetrically, the opposite end of the Transitive Axis governs the realization of the external argument. When the grammar must explicitly conceptualize a participant as the causal instigator or the starting point of the eventive trajectory, it deploys the spatial vector of Origin, or Reverse Inclusion ($\supseteq$). This operator physically anchors a peripheral participant to the event as its mereological source. This is the geometric reality underlying the so-called "Ergative" case, which explains its systematic cross-linguistic syncretism with the Instrumental (and the Ablative, cf. Franco and Manzini 2017). The Ergative is not a structural feature assigned by a functional v head; it is the materialization of the Origin vector ($\supseteq$) applied to an active initiator.

However, natural languages frequently allow the external argument to merge directly into the clause without the support of the ($\supseteq$) spatial relator, surfacing as an unmarked, caseless root (the so-called Nominative subject). Unlike the unmarked object, the direct merger of the subject does

not result in pseudo-incorporation. Because the internal variable of the core event is already saturated by the object, the unmarked subject is merged *in situ* as an adjunct at the outermost topological boundary of the eventive space (the highest segment of the VP). It is not absorbed into the eventive *mass*, but rather rests on its outer perimeter. It acts as the topological ground or locus of the transition without requiring an explicit adpositional vector. Consequently, the entire architecture of the transitive clause is resolved through spatial geometry: one bare root is absorbed within the core (the Object), another rests on its outer boundary (the Subject), and whenever these arguments require referential insulation or causal explicitness, the grammar erects the material boundaries of $\subseteq$ and $\supseteq$.

1.3. The Formalization of the Transitive Axis

Before turning to the empirical deconstruction of the *pure* structural object, it is necessary to explicitly formalize the theoretical machinery underlying our proposal. If the Minimalist Program (Chomsky 1995) is to be taken literally, the computational system should be stripped of all abstract functional hierarchies designed merely to satisfy internal derivational needs (e.g., vP , KP , αP).

As defined in the bare phrase structure framework, the core computational operation *Merge* takes two syntactic objects, α and β , and forms a new, hierarchically structured set $K = \{\gamma, \{\alpha, \beta\}\}$, where γ is the label of the projecting head. As we have seen, in contemporary generative accounts, the construction of a transitive clause requires multiple applications of *Merge* to project invisible functional heads (e.g., v , Voice, Appl), whose sole purpose is to assign uninterpretable Case features.

Under our strictly lexicalist perspective, the grammar is equipped only with substantive roots and elementary spatial relators, chiefly Zonal Inclusion ($\subseteq$) and Reverse Inclusion/Origin ($\supseteq$). The transitive clause is not a functional sequence of case-assigning heads, but a topological space constructed by the successive, costless merger of these elements.

When the Conceptual-Intentional (C-I) interface requires an internal argument to be highly salient (e.g., definite or animate), the syntax refuses to pseudo-incorporate it as a bare nominal root ($\emptyset$) into the event. Instead, the derivation of the Transitive Axis proceeds via *Merge* as follows:

(6) The Formal Derivation of the Transitive Axis via *Merge*

Step 1. Bounding the Object (DOM): The salient internal argument is merged with the Zonal Inclusion operator to prevent eventive absorption, forming a spatial boundary.

$\text{Merge}(\subseteq, DPOBJ) \rightarrow \{\subseteq, \{\subseteq, DPOBJ\}\}$

Step 2. Anchoring the Event: The eventive root (V) merges with the bounded object, projecting the event.

Merge $(V, \{\subseteq, \dots\}) \rightarrow \{V, \{V, \{\subseteq, \{\subseteq, DPOBJ\}\}\}\}$

Step 3. Bounding the Subject: The external argument can merge with the Reverse Inclusion operator to be defined as the mereological origin/source of the event.

Merge $(\supseteq, DPSUBJ) \rightarrow \{\supseteq, \{\supseteq, DPSUBJ\}\}$

Step 4. Closing the Axis: The spatially bounded origin merges with the event projection.

Merge $(\{\supseteq, \dots\}, \{V, \dots\}) \rightarrow \{V, \{\{\supseteq, \{\supseteq, DPSUBJ\}\}, \{V, \dots\}\}\}$

In this architecture, the assignment of a "Case" role (such as Accusative/DOM or Ergative) is nothing more than the direct outcome of merging a nominal root with a (spatial/relational) operator. The result is immediately legible at the C-I interface as the Goal/Possessor ($\subseteq$) or Origin ($\supseteq$) of the event, and at the Sensorimotor (S-M) interface as a specific preposition or case marker (e.g., Romance *a, pe*; Indo-Aryan *-ko, -ne*). No abstract [uCase] features are generated, and no "little- v " is required to license them.

Crucially, this *geometric* architecture predicts the entire typological space of morphosyntactic alignments without invoking stipulative parameters or invisible functional heads. An Ergative-Absolutive system emerges naturally when a given grammar tends only to bound the Source ($\supseteq$), leaving the Patient to pseudo-incorporate as a bare root ($\emptyset$). Conversely, a Nominative-(DOM) Accusative system arises when the syntax tend to bound the Goal ($\subseteq$) to prevent incorporation, while the Agent merges as a bare topological ground ($\emptyset$) at the perimeter. Most importantly, the model effortlessly derives Double Oblique and Tripartite systems, a theoretical problem for Dependent Case frameworks (cf. Atlamaz and Baker 2018). When a given grammar defines both participants are highly salient and resist fusion, the grammar simultaneously deploys both spatial operators: the Agent receives the origin vector ($\supseteq$) and the Patient receives the inclusion vector ($\subseteq$). Since the single argument of an intransitive verb (S) remains an unincorporated bare root ($\emptyset$) resting on the event's perimeter, the resulting $A (\supseteq) \neq P (\subseteq) \neq S (\emptyset)$ geometry materializes a Tripartite alignment.

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This radical elimination of uninterpretable features necessarily forces a redefinition of the operation *Agree*. As discussed in Section 1.1, standard Minimalism treats *Agree* as an asymmetrical,

distance-based operation whose sole purpose is the valuation and deletion of illegitimate features before Spell-Out. Conversely, Manzini and Savoia (2007, 2011) demonstrate that *Agree* can be reduced to a much simpler, interface-driven mechanism of "Minimal Search and Match". Building on the Pronominal Argument Hypothesis (Jelinek 1984), they propose that inflectional morphemes and clitics are not the spell-out of abstract agreement features on a functional head, but are themselves the true arguments of the verb, carrying a full referential burden. Consequently, all features involved in the derivation are interpretable.

What, then, is the role of *Agree*? It is simply the operation that establishes equivalence classes between coreferential bundles. When the syntax merges an oblique clitic ($\subseteq$) and a salient lexical DP (the DOM object, $\subseteq$) within the same domain, *Agree* does not delete any features. It merely certifies their featural identity, instructing the C-I interface to read the spatial relator and the nominal root as pointing to a single referent in the discourse. By reducing *Merge* to a topological set-formation of meaningful morphemes and *Agree* to the establishment of referential equivalence classes, the need for abstract structural Case ceases. As we will demonstrate in the next section, the so-called "pure" (unmarked) structural Accusative does not exist as an independent syntactic primitive.

2. 'Pure' Accusative as no case at all

2.1 Pseudo-incorporation and the Caseless Object

Within standard Minimalist framework, the licensing of the internal argument is strictly contingent on the assignment of structural Accusative Case by the functional head v (Chomsky 2000, 2001). This assumption, as we have already pointed out, entails that all direct objects, whether overtly marked by DOM or completely bare, must carry an uninterpretable [uCase] feature that makes them visible to the computational system. However, a growing body of cross-linguistic and theoretical evidence suggests that this premise is flawed: unmarked, non-specific objects are not elements whose Case feature has been valued as "null" or deleted at PF; they are entirely caseless (cf. Danon 2006).

If we adopt the geometric architecture outlined in Section 1.5, the "pure" (unmarked) direct object ceases to be an independent syntactic primitive. When an internal argument lacks specific referentiality, animacy, or topicality, the Conceptual-Intentional interface does not require it to be topologically bounded or extracted from the event. Consequently, the syntax does not merge it with the Zonal Inclusion operator ($\subseteq$). Instead, the object undergoes a process of direct topological

fusion with the verbal root, a phenomenon extensively formalized in the literature as *pseudo-incorporation* (Massam 2001; Dayal 2011; López 2012).

In pseudo-incorporation, the object is introduced as a bare, a-categorial nominal item ($\sqrt{N}$ or NP lacking the K-layer) that merely saturates the open argument place (the R-role, cf. Williams 1994) of the eventive predicate. Crucially, as demonstrated by Massam (2001) for Niuean, pseudo-incorporated objects do not trigger agreement, cannot undergo movement, and exhibit strict linear adjacency to the verb precisely because they form a complex predicate with it, rather than standing as independent, Case-bearing arguments.

(7) Niuean (Pseudo noun incorporation of the caseless object)

- a. Takafaga tūmau e ia e tau ika
 hunt always ERG he D PL fish
 ‘He is always hunting fish.’ (Independent, Case-marked object)

- b. Takafaga ika tūmau a ia
 hunt fish always D he
 ‘He is always fish-hunting.’ (Pseudo-incorporated, bare root) (Massam 2001: 157)

As (7b) illustrates, when the nominal root *ika* ('fish') is pseudo-incorporated, the clause undergoes a radical geometric restructuring. To avoid morphological confusion, we must analyze the markers transparently: in Niuean, the pronominal Ergative case marker *e* happens to be homophonous with the common noun article *e*, while the personal article (obligatory for bare pronouns and proper names) is *a* (Clark 1976, Massam 2001). In the fully transitive construction (7a), the pronoun *ia* ('he') is anchored to the event via the Ergative spatial vector *e* ($\exists$), while the object *tau ika* is a full DP headed by the common article *e*.

However, when the object *ika* pseudo-incorporates in (7b), it sheds its entire DP layer to fuse directly with the verbal core. Because the internal argument is no longer spatially individuated, the transitive geometry collapses. Consequently, the external argument is stripped of its Ergative spatial vector *e*. Reverting to a bare, caseless DP, the pronoun naturally surfaces with its default personal article *a* (*a ia*).

As Polinsky (2016: 130) notes, while pseudo-incorporation occurs in both Nominative-Accusative and Ergative languages, only in the latter is this dynamic "*morphologically visible, in the sense that the case of the subject changes from ergative to absolutive*" (i.e., detransitivization). If Ergative were a structural Case assigned by a functional ν head, it should not disappear merely because the object loses its D-layer.

The same mechanics apply to non-DOM objects in languages with differential marking. As convincingly argued by López (2012) for Romance, and extending to Indo-Aryan (Dayal 2011), the unmarked object is structurally deficient. It lacks the functional layer (KaseP) necessary to act as an independent referential anchor.

We propose that the syntax of the "pure accusative" is equivalent to the absence of the spatial relator ($\sqsubseteq$). The derivation of a caseless, pseudo-incorporated object proceeds via the direct merger of the nominal root with the eventive root, as formalized in (8):

- (8) Bare Object Fusion (Pseudo-incorporation)
 Merge ($\sqrt{V}, \sqrt{N_{OBJ}}$) $\rightarrow$ $\{\sqrt{V}, \{\sqrt{V}, \sqrt{N_{OBJ}}\}\}$

Because the nominal root $\sqrt{N_{OBJ}}$ in (8) is absorbed into the eventive space of V , it projects no spatial boundary. It has no uninterpretable features to check, and it does not enter into any *Agree* relation with a hypothetical v head. The morphological consequence of this architectural bareness is that the object receives a "zero" exponent.

The postulation of an abstract structural Accusative Case for these bare elements is born from a failure to distinguish between full spatial/relational arguments (DOM) and incorporated properties of the event (bare DP objects). If the unmarked object is caseless, the structural Case assignment by v becomes superfluous.

2.2 Latin: Neuters, Class Markers, and the Fall of the Accusative

A persistent objection to the elimination of the structural Accusative stems from the morphological richness of early Indo-European languages. Proponents of contemporary Minimalism frequently point to languages like Latin, Greek, or Russian, where the direct object of a transitive clause often exhibits an overt, dedicated morphological exponent (e.g., Latin *-m*, as in *domin-u-m*, *ros-a-m*). Within a standard feature-checking framework (Chomsky 2001), this overt ending is taken as the PF realization of an abstract [$u\text{Case: Acc}$] feature valued by the functional head v . However, a morphosyntactic and diachronic analysis reveals that the Indo-European "Accusative" is not a relational Case primitive, but a morphological device resulting from nominal classification.

The first empirical challenge to the standard *Agree* calculus comes from the pervasive phenomenon of *neuter* syncretism. Across the Indo-European family, nouns belonging to the neuter gender, which prototypically encode inanimate matter, masses, or abstract concepts, systematically exhibit a formal identity between the Nominative and the Accusative (Bossong 1991).

- (9) Latin Neuter Syncretism (Nom = Acc)
- a. Singular: Bell-u-m (war), mar-e (sea), flum-e-n (river)
 - b. Plural: bell-a (wars), mari-a (seas), flumin-a (rivers)

If Accusative Case were truly a structural feature assigned or valued by v in a specific syntactic configuration, one must ask why the computational system systematically fails to assign it, or deletes it at PF, precisely when the object denotes inanimate substance. We think that postulating zero-morphemes or post-syntactic Impoverishment rules (cf. Keine 2010) for all neuter objects is an *ad-hoc* stipulation.

Under our architecture, the neuter syncretism receives a principled explanation. As argued in Section 2.1, inanimate/mass objects often do not require spatial bounding; they pseudo-incorporate into the event. The Latin neuter is the morphological manifestation of a bare, a-categorical root ($\emptyset$). It receives no Accusative case because it does not project a spatial boundary ($\subseteq$).² The neuter Nominative and Accusative are identical because they are the exact same syntactic object: pure, unbounded lexical reference.

What, then, is the nature of the *-m* suffix found in Latin masculine and feminine objects? If it is not a relator or a structural Case, its presence must be accounted for. Drawing on work by Stark (2008), we argue that the Latin declension system operated fundamentally as a device for nominal classification within the universal dimension of apprehension. Latin nominal morphology did not merely mark syntactic dependency; it indicated whether an entity was a contoured/individualized object, a diffuse substance, or a collective (cf. Manzini and Savoia 2011, 2017, Franco et al. 2015, 2020).

This classification potential is visible in Latin roots that oscillate between genders, as demonstrated by Stark (2008):

- (10) Latin Nominal Classification via Gender/Number
- a. case-u-s (masc.) 'one single (piece of) cheese'

² Our idea of a radical caselessness of the unmarked object deals a fatal blow to the hierarchies of Nanosyntax (Caha 2009; Zompi 2019). Nanosyntax postulates a rigid, universal functional sequence (*f-seq*) in which marked and oblique cases are monotonically built by stacking functional K-heads on top of a foundational Nominative or Absolutive projection (e.g., *nom* $\subset$ *acc* $\subset$ *obl*). However, if the pure accusative object (much like the unmarked nominative) is not a feature bundle but a bare, a-categorical root ($\emptyset$) lacking any K-layer, the foundational node of this Russian-doll architecture is a syntactic phantom. Without a structural base, the entire sequence collapses. Furthermore, to salvage the strict linear contiguity of the *ABA generalization from the massive cross-linguistic syncretisms between DOM (Accusative) and Dative, Nanosyntactic approaches are forced to arbitrarily multiply functional projections, inventing phantom doublets like Acc1/Acc2, Dat1/Dat2, or Sacc/Bacc ("structural" vs "bigger" accusative). In our spatial-relational model, this massive proliferation of heads is entirely superfluous. The morphological identity between DOM and Dative is not an accidental overlap in a one-dimensional sequence, but the direct spell-out of the exact same elementary spatial predicate of Zonal Inclusion ($\subseteq$) erecting a boundary against pseudo-incorporation.

- b. case-u-m (neut.) 'cheese as a substance'
- c. frument-u-m (neut. sg.) 'wheat'
- d. frument-a (neut. pl.) 'corn (collective)' (adapted from Stark 2008)

Building on the lexicalist morphosyntax of Manzini and Savoia (2007, 2011), we formalize the Latin *-m* (and its Greek or Slavic counterparts) not as a relational K(ase) head, but as the intersection of an Infl(ection) node with a Class specification. When a highly individualized, animate, or contoured root (masculine/feminine) enters the transitive event space as a Theme, its internal morphological architecture requires a closure to signal its "contoured" nature, distinguishing it from a diffuse mass (namely, it is a class marker device). The *-m* ending is precisely this Class/Infl closer. It does not project a directional vector or a spatial boundary linking it to the verb; it merely *seals* the nominal phase.

A further proof that the Latin Accusative was a nominal class marker rather than a structural Case assigner is provided by diachrony. In the transition from Late Latin to Proto-Romance, the final *-m* was lost due to phonetic erosion. If the standard generative approach were correct, and the *-m* was the spell-out of an uninterpretable feature crucial for licensing the object, its disappearance should have triggered a systemic collapse, forcing the syntax to immediately recruit a dummy adposition to "save" bare objects. This did not happen. The "pure" direct object transitioned into Romance as a completely bare, morphologically zero element (*il libro, la casa*) without causing any syntactic crash, fulfilling its role by fusing with the verb.³ The only domain where Romance languages were forced to innovate an overt adpositional marker (Western Romance *a*, Romanian *pe*, etc.) was exactly where spatial bounding was conceptually obligatory: with highly salient, animate, and definite human referents. In other words, the true meaningful cases of Latin (the Dative ($\subseteq$) and the Ablative ($\supseteq$)) were preserved and replaced by prepositions to maintain the geometry of the event. The Accusative vanished into nothingness, because (relationally) it was already nothing, or better, it was a class morpheme.

Our view of a radical caselessness of the unmarked object finds a surprising, albeit partial, convergence within recent developments of Dependent Case Theory (DCT) itself. As demonstrated

³ It is interesting to note that true conceptual incorporation (idiomatization) is overwhelmingly restricted to the internal argument. As noted by Polinsky (2016), cross-linguistically there is a strong asymmetry between the abundance of Verb-Object idioms and the extreme rarity of Subject idioms (Manning 1996; Ura 2000). In our framework, this asymmetry follows naturally: the object can merge directly with the event core via pseudo-incorporation ($\emptyset$), whereas the external argument is a topologically independent Origin that resists eventive fusion. Furthermore, this explains the systematic incompatibility between idioms and Differential Object Marking (DOM). Because idiom chunks are non-referential by nature and do not make good topics, they intrinsically fail to trigger DOM, which is sensitive to high referentiality and topicality. Deploying the DOM spatial/relational boundary ($\subseteq$) would isolate the nominal root from the verb, thereby destroying the conceptual fusion required to yield the idiomatic reading.

by Kornfilt and Preminger (2015), phenomena such as raising-to-accusative in Sakha strictly require us to view the Nominative (and by theoretical extension, the Absolutive) not as a structurally assigned feature, but literally as "no case at all". For a nominal to successfully receive dependent Accusative upon raising into a higher phase, it must enter the new domain as a bare, caseless root. However, while Kornfilt and Preminger use this insight to salvage the configurational calculus of DCT, stipulating that the algorithm can only target caseless DPs to evaluate Case competition, we push their premise to its ultimate geometric conclusion. If the unmarked argument is demonstrably a topological baseline devoid of abstract Case features, the necessity for a purely structural algorithm to calculate "marked" cases completely evaporates. Overt morphological markings such as the Accusative (DOM) or the Ergative are not the mechanical output of asymmetric c-command relations computed at Spell-Out. Rather, they are the direct syntactic insertion of explicit, semantically contentful spatial relators (such as Zonal Inclusion $\subseteq$ or Origin $\ni$) designed to anchor these otherwise bare, caseless roots to the topology of the event.

2.3 On Burzio's Generalization: *Smuggling and Geometric Independence*

The realization that the "pure" unmarked object is a caseless, pseudo-incorporated root logically necessitates the dismantling of one of the most entrenched axioms of generative syntax: Burzio's Generalization (Burzio 1986). In its standard formulation, Burzio's Generalization posits a strict bidirectional dependency between the assignment of an external θ -role and the availability of structural Accusative case. The canonical passive voice is consequently treated as a derivational operation: the suppression of the external argument "absorbs" the Accusative case, forcing the caseless internal argument to undergo A-movement to Spec,TP to receive Nominative case and satisfy the Case Filter.

In recent Minimalist attempts to maintain this derivational link while accommodating strict locality conditions (Relativized Minimality, Rizzi 1990), standard accounts have resorted to increasingly *baroque* architectures. The culmination of this trend is the "smuggling" analysis proposed by Collins (2005). To explain how the internal argument can move to the subject position without illegally crossing the external argument merged in Spec,vP, Collins posits that the entire VP moves to Spec,VoiceP, effectively "smuggling" the object past the agent. Crucially, this approach requires treating the preposition introducing the agent (e.g., English *by*, Italian *da*) as a "dummy" functional head devoid of semantic content, existing solely to satisfy structural requirements.

As demonstrated by Manzini (2017) and Franco et al. (2021), the smuggling architecture relies on semantically vacuous movement operations that are illegible at the Conceptual-Intentional

interface, and it violates the basic requirement that prepositional heads must bear interpretable content. Smuggling is motivated by the need to save the (the postulation of) structural Accusative case and its absorption.

If we adopt the spatial-relational architecture where the unmarked object is inherently caseless ($\emptyset$), the entire premise of Burzio's Generalization evaporates. There is no structural Accusative case to be absorbed, and consequently, the passive cannot be a derivation from the active. Active and passive are not derivationally linked; they are two independent architectures for event construction.

In a passive configuration, the internal argument saturates the eventive space exactly as it does in the active voice: by merging as a bare nominal root. Because the passive (perfect) participle lacks the active functional layer required to natively host an external argument, the syntax cannot merge an agent directly onto the clausal spine. Instead, the grammar recruits the spatial/relational lexicon. To express the agent, the syntax merges a nominal root with the Reverse Inclusion/Origin operator ($\supseteq$), formatting the participant as the mereological source of the causation event. This oblique constituent is then directly adjoined to the VP. Consider the structure in (11).

(11) The Geometry of the Passive (as an Independent Architecture)

- a. La casa è costruita dal muratore.
the house is built by.the mason
- b. [VP [VP costruita la casa] [P($\supseteq$) da-l muratore]]

In (11b), the preposition *da* is not a "dummy" head, and no VP-smuggling or abstract *vCAUSE* node is required. In perfect adherence to the spatial geometry of the Transitive Axis depicted in (6), the eventive core merges directly with the caseless nominal root (*la casa*), forming the bare VP. To introduce the agent, the syntax recruits the spatial lexicon: it merges a nominal root with the Reverse Inclusion/Origin operator ($\supseteq$), formatting the participant as the mereological source of the causation event. This oblique constituent is then directly adjoined to the eventive layer (VP). *Da* is an active, elementary spatial predicate ($\supseteq$) that explicitly instructs the interface to interpret *the mason* as the origin or author of the event, rendering any invisible functional projection of causation redundant. In our framework, the verb *to be* (Italian *essere*) is not a mere auxiliary, but the lexicalization of the higher eventive head (T or CAUSE) that anchors the resultant state within the temporal domain of the clause. Within the geometric blueprint of the passive (11b), *essere* occupies the apical position of the eventive spine, above for the resultative VP core.

By merging at this higher level, *essere* instructs the interface to interpret the entire configuration as a "witnessed" or "subsisting" state. Crucially, we argue that this head is stripped of abstract Case-valuing features, and the movement of the internal argument (*la casa*) to the preverbal Spec,TP position is entirely disconnected from Case valuation. Instead, this raising is driven by interface requirements such as the Extended Projection Principle (EPP) or Topic/aboutness conditions.

Thus, we assume that Burzio's Generalization attempts to establish a causal link between two syntactic *phantoms*: a non-existent structural Accusative feature and a non-existent abstract functional specifier for the Agent. Once transitivity is recognized as the simple (optional) topological adjunction of spatial operators ($\subseteq$ and $\supseteq$) to an event, the need for absorption, smuggling, and dummy prepositions vanishes completely.⁴

The spatial-relational formalization of the Agent as an oblique Origin vector ($\supseteq$) provides a possible solution to the problem of argument optionality. In standard generative accounts, the transitive axis is strictly dependent on the projection of the external argument: if the Agent is suppressed or demoted (as in passives), the functional head *v* is rendered "defective" and loses its ability to assign Accusative case to the object. Consequently, the object is forced to move to a higher functional specifier to satisfy the Case Filter. As previously argued, this derivational link between active and passive voices is a theoretical construct and, possibly, not empirical reality. Under the WYSIWYG principle, the core of a transitive clause (e.g., the VP containing the internal argument) is geometrically complete in itself. The internal argument merges directly as a bare, caseless root representing the locus of the state or transition. The Agent, conversely, is not a structural necessity forced by a functional probe, but rather a peripheral spatial participant anchored to the event via the Reverse Inclusion operator ($\supseteq$).

⁴ The differential behaviour of marked objects under passivization has long been used by Case theorists against spatial or oblique analyses of DOM. As extensively documented by Bányi (2018), while true Dative arguments resist periphrastic passivization, DOM objects easily promote to subjecthood, losing their overt marker in the process (e.g., Spanish *veo a la mujer* 'I see DOM the woman' vs. *la mujer fue vista* 'the woman was seen'). In standard Minimalist accounts, this morphological disappearance is taken as proof that DOM is simply an abstract structural Accusative assigned by *v*. Within our spatial-relational framework, this contrast requires no appeal to invisible functional heads or abstract Case-feature deletion. It follows directly from the radical topological difference between active and passive configurations. In the active transitive axis, a highly individuated internal argument requires the explicit deployment of the Zonal Inclusion operator ($\subseteq$) to avoid conceptual absorption into the eventive core. In the passive configuration, however, the transitive axis is dismantled: the external argument is demoted to an adjunct Origin ($\supseteq$), and the active causative vector collapses. Stripped of the active eventive tension, the internal argument is no longer *in danger of* pseudo-incorporation. Consequently, the necessity for the spatial boundary ($\subseteq$) completely ceases. The internal argument emerges as what it fundamentally is: a bare, caseless topological ground ($\emptyset$) that naturally surfaces as an unmarked subject. Thus, the disappearance of DOM under passivization is not the spell-out of an abstract Case feature, but the literal geometric dismantling of a spatial boundary when the active topological frame ceases to exist.

Given that the ($\supseteq$) relator operates topologically as an oblique adjunct (whether it surfaces as the preposition *da* in the Italian passive, or as an Ergative case marker), its projection is inherently optional. When the speaker wishes to explicitly identify the active initiator of the causal chain, the ($\supseteq$) vector is merged. When the source of the event is irrelevant, unknown, or generic, the syntax simply opts not to project the spatial adjunct, leaving the origin of the event existentially closed at the Conceptual-Intentional interface. This explains why the *by*-phrase (*da-l muratore*) can be freely omitted in passive architectures without violating any syntactic constraints. The "transitive axis" is fully preserved because the event topology (the resultant state of the patient) does not depend on the overt materialization of its starting point.

This optionality extends to ergative alignments. If the Ergative case is the morphological spell-out of the ($\supseteq$) adjunct, we predict that ergative agents should be freely omissible without triggering a valency-reducing operation (like an antipassive) and without stripping the object of its Absolutive status. This prediction is confirmed by typological data. In Lezgian, for instance, the Ergative agent can undergo free omission while leaving the transitive valence pattern fully intact (Haspelmath 1993). Similarly, in the Amazonian isolate Trumai, the Ergative argument is frequently and voluntarily suppressed for discourse-pragmatic reasons (e.g., to promote the Absolutive argument as the sole accessible antecedent for anaphora across clauses). Consider the example in (12).

(12) Trumai (Omission of the Ergative Agent without detransitivization)

dinoxoxo	tete-kma-s	pita-n.
girl(ABS)	body.paint-PRF-TEMP	go.out-3ABS

‘When (someone) finished painting the girl, she left.’ (Guirardello-Damian 2010: 207)

In (12), the suppression of the Ergative agent does not crash the derivation, nor does it alter the Absolutive status of the patient (*dinoxoxo*). The event of "body-painting" retains its transitive geometry. The syntax merely declines to merge the ($\supseteq$) relator, proving that the Agent is an optional, modular spatial addition to the eventive core. Transitivity is thus an intrinsic property of the event structure, not a mechanical byproduct of the presence of an external argument.

The topological optionality of the Agent as an external Origin vector ($\supseteq$) raises a natural architectural question regarding the internal argument. If the external argument can be freely omitted, why is the internal argument geometrically obligatory, and how does this asymmetry account for the phenomenon of "null objects"?

In standard generative grammar, following the seminal work of Rizzi (1986), it is widely recognized that languages like Italian allow for syntactically active but phonologically silent objects, formalized as arbitrary *pro* (e.g., *Questa musica riconcilia [pro] con se stessi* 'This music reconciles [one] with oneself'). The literature generally assumes a structural distinction between syntactically projected null pronouns (*pro*) and lexically saturated implicit arguments. Under our spatial-relational architecture, this asymmetry is not a problem, but a geometric consequence.

While the Agent is a peripheral starting point of the causal chain (a spatial adjunct $\supseteq$ that can be left unspecified), the internal argument constitutes the absolute topological Locus of the eventive transition (the VP core). The eventive space must be saturated, either by an overt nominal root, an implicit argument, or a syntactically projected silent pronoun (*pro*).

The existence of silent objects perfectly validates the Anti-Incorporation rationale of Differential Object Marking. As Rizzi (1986) demonstrated, Italian null objects are strictly limited to an arbitrary, generic interpretation ([+human, +generic]). Because these silent referents lack specific individuation and discourse prominence, they offer zero topological resistance to the verbal root. They smoothly undergo pseudo-incorporation into the eventive core without requiring spatial shielding. The Zonal Inclusion operator ($\subseteq$, the DOM boundary) is deployed by the syntax only when an internal argument crosses the threshold of specific referentiality, demanding physical separation from the predicate.⁵ A silent, arbitrary *pro* saturates the eventive locus without triggering the deployment of an overt topological boundary.

2.4 Radical Caselessness: Evidence from Ergative Alignments

The radical conclusion that the "pure" direct object is a syntactically caseless root ($\emptyset$) finds its empirical validation in the morphosyntactic alignments of ergative languages. If Agree approaches

⁵ The spatial-relational framework also provides a principled account for the problematic assignment of Ergative case to the subjects of unergative verbs, as well as the cross-linguistic variability associated with it (Fluid-S or Active alignments). Following the seminal insights of Hale & Keyser (1993), unergative predicates are best analyzed as *concealed transitives* resulting from the incorporation of a nominal root into a light verb (e.g., *sleep* as *do a sleep*). This incorporated nominal root acts as the structural Locus of the event, saturating the internal space of the transition. Consequently, the single argument of an unergative verb (unlike that of an unaccusative) must be introduced externally to the event core via the Origin vector ($\supseteq$), correctly predicting the widespread appearance of Ergative case on unergative subjects (e.g., in Basque or Georgian). Given that the ($\supseteq$) relator is a peripheral spatial adjunct rather than a rigid structural requirement, our model captures the phenomenon of "optional" or "fluid" ergativity on intransitives without postulating *ad hoc* lexical features. As extensively documented, in languages like Hindi/Urdu or the Nakh-Daghestanian language Tsova-Tush (see Holisky 1987), the use of the ergative marker on unergative verbs (e.g., *cough*, *bark*, etc.) is subject to speaker choice, typically correlating with volitionality. From a geometric perspective, this variability is the morphological reflex of topological optionality: if the speaker construes the participant as the active, volitional initiator of the event chain, the ($\supseteq$) boundary is overtly merged. If the event is construed as an accidental or non-volitional happening, the speaker opts not to project the spatial adjunct. Thus, fluid ergativity is not a breakdown of Case algorithms, but the natural consequence of manipulating spatial vectors at the conceptual-intentional interface.

and Dependent Case Theories (DCT) were correct in assuming that all nominals require structural licensing, the Absolutive case, which typically marks both the intransitive subject (S) and the transitive object (O), should be analyzable as the morphological spell-out of an abstract [uCase] feature assigned by a functional head (such as T or ν). However, the cross-linguistic record refutes this assumption.

As extensively documented in typological surveys, the Absolutive case in ergative systems almost always has zero realisation or at least a zero allomorph. This zero-marking is not merely a phonological accident or a generalized spell-out rule for default features, but rather the direct morphological reflection of syntactic bareness. Consider the canonical ergative-absolutive paradigm of Dyirbal (Pama-Nyungan), where common nouns exhibit overt morphology only when functioning as transitive agents.

(13) Dyirbal (Zero-marked Absolutive as a bare root)

- a. yabu- $\emptyset$ banaga-nyu
 mother-ABS return-NONFUT
 ‘Mother (S) returned.’
- b. nguma-nggu yabu- $\emptyset$ bura-n
 father-ERG mother-ABS see-NONFUT
 ‘Father (A) saw mother (O).’ (Dixon 1994: 14)

In both (13a) and (13b), the nominal *yabu* (‘mother’) surfaces as a bare, uninflected root. In the standard generative calculus, it seems that one is forced to stipulate that in (13a) the invisible Absolutive case is assigned by T, whereas in (13b) the exact same invisible Absolutive case is assigned by ν or by a dependent case calculus. This leads to an architectural redundancy: the grammar supposedly performs complex feature-valuation operations multiple times solely to assign a "zero" exponent. Under our spatial-relational model, the explanation is geometrically transparent: *yabu* receives no morphological case simply because it has no syntactic Case. It lacks the spatial boundary operator ($\subseteq$ or $\supseteq$) and merely pseudo-incorporates into the event to saturate its open argument position.

The postulation of structural object Case is further dispelled by Pama-Nyungan languages exhibiting complex split systems, such as Djapu. Djapu is frequently analyzed as possessing a three-way case system (Ergative, Accusative, and Nominative/Absolutive), where human and animate transitive objects receive an overt Accusative suffix (*-nha* / *-n*), while lexical transitive subjects receive an Ergative suffix (*-y'* / *-dhu*). Crucially, however, inanimate and non-human

transitive objects systematically lack overt case morphology, surfacing with a zero exponent exactly like intransitive subjects.

(14) Djapu (Split object marking and Caselessness)

- a. Transitive with human object (DOM as Spatial Boundary $\subseteq$):

Mak rlinygu-n galka-y' ba:pa-'ngali-n dharpu-ngal.
 Maybe already-IM sorcerer-ERG father-KINPROP-ACC spear-PERF
 'Maybe a sorcerer has already speared your father.' (Morphy 1983: 111)

- b. Transitive with non-human object (Caseless bare root $\emptyset$):

Mirlipi- $\emptyset$ ngayi yurru dharpu-ma-n garapa-y.
 shoulder.blade-ABS he.NOM FUT pierce-UNM-IM spear-INSTR
 'He pierces the shoulder blade with a one-pronged spear.' (Morphy 1983: 39)

In recent accounts (e.g., Müller & Thomas 2015), the split pattern in Djapu requires stipulating complex, post-syntactic impoverishment rules that arbitrarily delete an abstract [Accusative] feature before Spell-Out just because the object is inanimate. Furthermore, the combination in (14b) poses a clear problem for Dependent Case Theory (Marantz 1991; Baker 2015). In (14b), a fully transitive event surfaces with two morphologically unmarked arguments (the Nominative pronoun *ngayi* and the inanimate *mirlipi*). If Case were a structural competition assigning a dependent feature whenever two caseless DPs occupy the same local domain, this double-zero spell-out should be impossible.

Djapu simply exhibits Differential Object Marking within an ergative alignment: the overt object suffix in (14a) is the Zonal Inclusion operator ($\subseteq$) deployed locally to spatially bound highly individuated (human) referents. Inanimate objects, as vividly illustrated in (13b), do not undergo "case-feature deletion", nor do they trigger dependent algorithms; they are simply bare, caseless roots ($\emptyset$) that saturate the eventive space without projecting a topological boundary. Similarly, a highly salient deictic center like a 1,2 pronoun functions as the Origin without requiring an overt Ergative boundary.

This typological data mirrors the Indo-European neuter syncretism analyzed in Section 2.2. The Latin neuter object *bell-u-m* (Nom=Acc) and the Australian absolutive object *yabu- $\emptyset$* (S=O) are not morphological anomalies or instances of "null Case". They are the exact same geometric object: a bare nominal root ($\sqrt{N}$) stripped of any spatial or relational boundary.

The cross-linguistic evidence converges on a single, radical conclusion: the "pure" structural object case does not exist. Transitivity is a spatial axis where arguments are either topologically

bounded by overt relational operators ($\subseteq$ for DOM/Dative, $\supseteq$ for Ergative/Ablative) or left as caseless roots ($\emptyset$) that fuse with the event.

3. Differential Object Marking as Zonal Inclusion ($\subseteq$)

3.1 *The Geometry of Salience (Anti-incorporation)*

Having established in Chapter 2 that the so-called "pure" accusative is reducible to a caseless, a-categorical root that pseudo-incorporates into the eventive core, we are now positioned to correctly identify the true nature of Differential Object Marking (DOM). The question that a minimalist architecture must answer is no longer "why do some objects receive extra Case?", but rather, "why do certain objects resist topological fusion with the verb?"

The cross-linguistic record demonstrates that DOM is sensitive to specific semantic properties of the internal argument, classically formalized as prominence hierarchies or scales. As noted in seminal works (Silverstein 1976; Comrie 1979; Bossong 1985, 1991; Aissen 2003), differential marking preferentially targets entities at the higher end of the Animacy and Definiteness/Specificity scales (e.g., 1st/2nd person pronouns > 3rd person pronouns > proper names > human > animate > inanimate).

As we have already seen, contemporary works in the generative frameworks have attempted to capture these hierarchy effects by postulating that salient nominals possess highly articulated functional structures (e.g., additional D or K layers) that bear uninterpretable Case features requiring licensing by a designated functional head, such as v or an *ad-hoc* α projection (Massam 2001; Danon 2006; Rodríguez-Mondoñedo 2007; López 2012; Ormazabal & Romero 2013; Kalin 2018). We think that translating semantic scales into abstract functional cartographies merely redescribes the phenomenon without providing explanatory adequacy.

From our perspective, the behavior of salient objects is driven by a much deeper cognitive and structural necessity: the resistance to eventive absorption(anti-incorporation. Highly individuated referents, such as speech act participants (1st and 2nd person pronouns), human beings, or strictly definite entities, possess intrinsic referential contours and deictic independence (Manzini & Franco 2016). Conceptually, a human agent or a deictic center cannot be easily conceptualized as an inert, undifferentiated property of a verbal action. Syntactically, this translates into an absolute ban on pseudo-incorporation. To remain legible at the Conceptual-Intentional (C-I) interface as an autonomous entity, the salient argument must be explicitly "extracted" from the core eventive space and shielded.

This necessity for ‘geometric extraction’ is perfectly observable in Romance languages exhibiting strict DOM on highly salient arguments, such as Romanian and Sardinian.

(15) Obligatory Anti-incorporation of Specific Humans

- a. L-am întâlnit *(pe) Ion.
 him-have.1SG met DOM John
 'I met John.' (Romanian; Mardale 2008)
- b. Appo vistu solu *(a) isse.
 have.1SG seen only DOM him
 'I saw only him.' (Sardinian; Mardale 2008)

The geometric nature of this anti-incorporation process is further corroborated by micro-variation in Southern Italian dialects. As documented by Ledgeway (2023), in the Abruzzese dialect of Arielli, the assignment of DOM is rigidly restricted to the absolute peak of the deictic scale: the 1st and 2nd person (see Manzini et al. 2020).

(16) Deictic Independence and the Person Split

- a. Seme viste a tte / a nnu.
 be.1PL seen DOM you / DOM us
 'We saw you / us.'
- b. Seme viste (*a) quille / (*a) Marije.
 be.1PL seen DOM that.one / DOM Maria
 'We saw that one / Maria.'
 (Arielli, Abruzzo; Ledgeway 2023)

In (16), the spatial boundary (*a*) is mandatory for Speech Act Participants because they are inherent deictic centers that anchor the event; they resist topological fusion. Conversely, 3rd person referents and proper names, lacking this absolute deictic anchoring, are treated as regular themes and forced to incorporate as bare roots (*quille*, *Marije*).

This geometric pattern is mirrored at the very bottom of the animacy scale. While a salient human referent cannot "melt" into the eventive mass, an inanimate root effortlessly saturates the internal argument slot by topologically fusing with the predicate. Consequently, the syntax rigorously forbids the insertion of a spatial boundary for inanimates.

(17) Bare Roots and Eventive Fusion (Bottom of the scale)

Appo vistu (*a) sa makkina.

have.1SG seen DOM the car

'I saw the car.' (Sardinian; Mardale 2008)

Crucially, in a minimalist, relational architecture devoid of functional heads, the syntactic component cannot execute the extraction of salient arguments (14-15) by assigning a [uCase] feature or triggering semantically vacuous A-movement. The only mechanism the grammar possesses to prevent the fusion of two roots is the interposition of a geometric boundary. The salient argument must be wrapped in an elementary relator that provides it with a distinct topological coordinate.

Differential Object Marking is precisely the overt morphological manifestation of this spatial bounding. Because the object is too prominent to be a mere internal property of the verb, the syntax must construct an asymmetric spatial relation, extracting the referent and connecting it to the event from the outside. As we will demonstrate in the next section, the specific topological operator universally recruited by natural languages to fulfill this bounding function for internal arguments is the elementary predicate of Zonal Inclusion ($\subseteq$).

A further crucial point to consider is the following. The conceptual and syntactic proximity of possession and location is a well-established universal in generative grammar, dating back to the foundational insights of Lyons (1967) and Freeze (1992). Early formal approaches correctly identified this isomorphism, famously proposing that possessive configurations are syntactically derived from an underlying locative template (Freeze 1992; den Dikken 1995). While we fully endorse their intuition regarding the deep structural unity of these domains, our spatial-relational architecture, in line with more recent typological and formal assessments (Levinson 2011; Myler 2016), effectively reverses this perspective. In our model, physical location is not the ultimate syntactic primitive from which possession is derived. Rather, both possession and spatial location are epiphenomenal manifestations of a more fundamental, abstract topological operator: Zonal Inclusion ($\subseteq$). Under our framework, a locative adposition or case is simply the elementary inclusion predicate ($\subseteq$) subjected to a specific spatial or geographical restriction (e.g., 'x is included by y, where y is a location', cf. Franco et al. 2021). By taking the part-whole topological relation as the absolute primitive, we naturally explain the pervasive cross-linguistic tendency to convey non-local relationships (such as possession, and eventually, clausal transitivity) utilizing spatial case morphology.

3.2 The Topological Operator $\subseteq$ and the Dative Syncretism

As we have pointed out in the previous section, in an architecture devoid of abstract functional heads, the extraction of a salient argument from the eventive core (anti-incorporation) cannot be executed by assigning a [uCase] feature. It must be executed by the spatial lexicon. To prevent topological fusion, the grammar deploys a relational primitive to bound the object. We propose that the specific operator universally recruited for this task is the elementary predicate of Zonal Inclusion ($\subseteq$).

When merged with a nominal root, the inclusion operator ($\subseteq$) creates a spatial domain that encompasses the referent. At the Conceptual-Intentional (C-I) interface, this geometry is interpreted as the topological terminus of the event, the Goal, the Experiencer, or the Possessor. Consequently, Differential Object Marking is not a structural "Case" triggered by an abstract αP head or an expanded v probe. It is literally the spatial mapping of a salient entity as the endpoint of the transitive vector.

This topological analysis provides a principled explanation for one of the most pervasive phenomena in the morphosyntax of case: the systematic syncretism between DOM, the Dative case, and the Allative adposition. Across numerous unrelated families, from Romance (*a*) to Indo-Aryan (Hindi *-ko*) and beyond, the morphology used to mark a specific/animate direct object is perfectly isomorphic to the morphology introducing the indirect object and/or the directional goal.

This isomorphism is problematic for Agree. Because Dative is considered an "inherent/lexical" case assigned by an Applicative head (cf. Cuervo 2003, Pytkänen 2002), while DOM is considered a "structural" case assigned by v , standard accounts are forced to dismiss their morphological identity as massive accidental homophony, or to stipulate the post-syntactic insertion of a "dummy" preposition to salvage a Caseless nominal.

However, as argued by Manzini and Franco (2016), neither adpositions nor applicatives (cf. Franco 2024) are dummy elements; they are both overt morphological exponents of the elementary relator $\subseteq$. The syncretism is not an accident, but a geometric necessity.

The fundamentally *topological* nature of DOM is further confirmed by the instability of DOM markers in language contact and diachrony, where speakers effortlessly replace traditional DOM markers with overt locative prepositions (and *vice versa*). In colloquial and diasporic Romanian,⁶ for instance, the standard DOM marker *pe* is frequently replaced by the locative/allative preposition *la* 'at/to', exactly because both satisfy the need for a $\subseteq$ boundary:

⁶ As Manzini and Franco (2016) point out, DOM is cross-linguistically syncretic with spatial and oblique relators. However, the Romanian marker *pe* presents an exceptionally transparent case of spatial geometry. Deriving from the Latin perlativ/directional *per*, *pe* functions synchronically as a fully-fledged locative preposition meaning 'on' (Hill &

(18) Locative Substitution of DOM (Romanian)

Când mai mergi strigă și pe / la noi.
when again go.2SG call.IMP.2SG and DOM / LOC 1PL

'Give us a shout when you go there again.'

(Mardale & Karatsareas 2020: 8)

The syntactic structure of the differentially marked object is thus identical to the structure of an indirect object. While a bare, pseudo-incorporated object involves the direct merger of the nominal root with the verb, the derivation of DOM requires the prior construction of a prepositional/spatial phase.

(19) The Geometry of DOM (Zonal Inclusion) Merge ($\subseteq, \sqrt{N}$) $\rightarrow \{\subseteq, \{\subseteq, \sqrt{N}\}\}$

The salient object receives Dative/Allative morphology simply because it *is* a topological Goal. It is extracted from the event and positioned as the domain into which the action resolves.

3.3 Reframing "Topic" as an Epiphenomenon

The geometric formalization of Differential Object Marking as Zonal Inclusion ($\subseteq$) allows us to resolve the long-standing debate regarding the role of information structure in argument encoding. A vast and highly influential body of typological and functionalist literature has established a robust cross-linguistic correlation between DOM and topicality. As extensively documented, differentially marked objects exhibit strong tendencies to be discourse-linked, highly individuated, and referentially persistent.⁷

Mardale 2021). Unlike the Spanish DOM marker *a*, *pe* is disconnected from the dative case, which Romanian marks inflectionally (Irimia 2020; Fábregas 2013, among others). The recruitment of a pure locative adposition to encode highly individuated direct objects provides direct morphological evidence that DOM is the projection of a topological boundary ($\subseteq$), rather than the spell-out of an abstract structural Case. This spatial and oblique nature of DOM extends robustly to the Indo-Iranian family. The Modern Persian DOM enclitic *-râ* / *-ro*, for instance, grammaticalized directly from an ancient postposition with allative/benefactive (dative) and goal semantics (Haig 2008). While Modern Persian has split these functions (retaining *-râ* for DOM and utilizing the preposition *be* for the Dative/Allative), other Iranian varieties systematically preserve the geometric syncretism. In Mazandarani, for example, the identical spatial postposition *-ra* / *-re* overtly lexicalizes both the Dative and the DOM (Lecoq 1989).

⁷ Notice that in the functional-typological literature, the emergence of DOM is frequently correlated not only with animacy and topicality, but also with the parameter of *affectedness* (Hopper & Thompson 1980; Naess 2004; von Heusinger & Kaiser 2011). While functional accounts debate the precise alignment of affectedness hierarchies (Tsunoda 1985; Beavers 2011) with object marking, our geometric framework captures this parameter naturally without treating it as an independent, primitive semantic feature. In topological terms, 'affectedness' translates to the degree of conceptual and spatial absorption of the internal argument into the eventive core. Highly affected objects undergo direct eventive fusion via pseudo-incorporation (surfacing as a caseless $\emptyset$ root). Conversely, when an object is 'unaffected', such as the stimuli of perception verbs or object-experiencer psych-verbs (cf. Belletti and Rizzi 1988), which, as von

Building on these observations, scholars have proposed that DOM fundamentally functions as a topic-marking device. In a seminal diachronic analysis, Iemmolo (2010) argues that DOM systematically emerges from the grammaticalization of left-dislocated topics, particularly personal pronouns, eventually spreading to other highly topical referents. In a parallel theoretical vein, Dalrymple and Nikolaeva (2011) propose that the primary synchronic function of DOM is to encode the "secondary topic" of a clause, a referent that is highly salient and stands in a specific "aboutness" relation to the primary topic (the subject).

While these functionalist approaches are descriptively accurate in capturing the discourse profile of marked objects, they commit a fundamental epistemological error: they confuse the *pragmatic effect* with the *syntactic cause*. Postulating "topicality" as the generative trigger for morphological case marking requires directly hardwiring pragmatic intentions (such as "aboutness" or "givenness") into narrow syntax. Cartographic approaches attempt to solve this by projecting dedicated TopP/FocP heads in the low periphery of the clause (cf. Belletti 2004), endowing the object with abstract [*uTopic*] features that must be checked by functional probes. However, as noted in recent critical re-evaluations (cf. Onea & Mardale 2020), this *direct* syntacticization of discourse violates the autonomy of the computational system: Narrow Syntax is solely an engine of structure-building; it does not "know" what a discourse topic is, nor does it manipulate communicative intentions. In our view, it manipulates only roots and geometries (cf. Nordström 2017).

If we adopt the spatial-relational architecture, the connection between DOM and topicality is instantly demystified: "Topic" is an epiphenomenon of geometric extraction. As argued in Section 3.1, when the syntax operates on a highly ranked referent (a deictic center, a human, or a definite entity), it encounters a geometric incompatibility: the entity is too highly individuated to be topologically fused (pseudo-incorporated) into the eventive core. To resolve this, the grammar deploys the Inclusion operator ($\subseteq$) to extract the nominal root and provide it with a distinct spatial boundary.

Crucially, it is this purely geometric operation of anti-incorporation that *generates* the pragmatic reading. When the derivation is transferred to the Conceptual-Intentional (C-I) interface, the interface reads an eventive core ($\sqrt{V}$) alongside a spatially independent, bounded nominal domain ($\subseteq P$). Because the object is topologically separated from the predicate rather than absorbed by it, the C-I interface interprets it as a referentially autonomous anchor. In the realm of discourse

Heusinger and Kaiser (2011) rigorously observe, strongly favor DOM with human referents, it intrinsically resists topological absorption. Consequently, the syntax is forced to deploy the Zonal Inclusion operator ($\subseteq$) to spatially bound and anchor the argument, preventing its fusion into the event core.

pragmatics, a referentially autonomous entity that anchors an event without being an active agent is exactly what functionalists describe as a "Topic" or "Secondary Topic".

Therefore, Iemmolo (2010) and Dalrymple & Nikolaeva (2011) observe the correct empirical reality, but reverse the causal arrow. The syntax does not insert a differential marker *because* the speaker wishes to topicalize the object. Rather, the object receives a "topic" interpretation at the C-I interface *because* the syntax was forced to wrap it in the spatial relator ($\subseteq$) to prevent illicit eventive fusion.

By reframing topicality as the interface legibility of the $\subseteq$ operator, we strip DOM of all functionalist teleology. Differential Object Marking is not a discourse strategy; it is the application of spatial topology, whose consequence is interpreted as discourse prominence.

4. The WYSIWYG Principle across Languages

Building on (the already discussed) Kornfilt and Preminger's insight that the unmarked Nominative is not a Case but rather a caseless bare root, our geometric model drives this logic to its natural conclusion: the structural Accusative does not exist. According to the WYSIWYG (*What You See Is What You Get*) principle, Differential Object Marking and accusative suffixes are never the spell-out of uninterpretable formal features assigned by a functional head.⁸ Instead, they are the literal materialization of the relators recruited by the syntax to enforce the *Anti-Incorporation Ban*. When an internal argument is highly salient, it resists fusion with the eventive core. To shield this referent, the grammar physically encapsulates it within a topological boundary, usually deploying the vector of Zonal Inclusion ($\subseteq$) or an absolute deictic pointer. This morphosyntactic reality renders the algebraic calculations of Dependent Case Theory entirely obsolete. The overt externalization of these spatial vectors is transparent in languages standardly invoked to support abstract case probes, as we will show in what follows.

⁸ Proponents of the structural Case approach frequently cite Dravidian languages, such as Kannada, where the differential object marker *-annu* is morphologically distinct from the Dative *-ge*, arguing that DOM is the realization of an abstract [Accusative] feature (e.g., Lidz 2006). However, a geometric analysis of Kannada strictly refutes this view, perfectly satisfying our Anti-Incorporation and Etymological Bans. Syntactically, *-annu* acts solely as a topological shield: while animate objects obligatorily require it to prevent eventive fusion, inanimate objects strictly alternate between bare pseudo-incorporation (yielding an indefinite reading) and *-annu* marking (forcing a definite, extracted reading, cf. Steever 1998: 20). Morphologically, the marker *-annu* does not emerge *ex nihilo*. Within the Dravidian spatial system, the base *a-* is the universal deictic root for distal orientation, serving as the foundation for both the Locative adposition (*-alli*) and the DOM marker (*-annu*) (Steever 1998: 134). By wrapping the salient object in a deictic-spatial pointer, Kannada syntax topologically separates the referent from the verb. Thus, even when DOM diverges from the Dative, it remains fundamentally anchored in the geometry of spatial location.

4.1 The postulation of Abstract Accusative and Adpositional Evidence

4.1.1 Paraguayan Guaraní: Locatives, Possessors, and the Inactive Paradigm

Paraguayan Guaraní (Tupí-Guaraní) provides interesting empirical evidence. Formal approaches assume that Guaraní alignment and Differential Object Marking (DOM) involve complex functional ϕ -probes or abstract Case assignment (see Zubizarreta and Pancheva 2017). However, looking at the morphological substance reveals a purely geometric reality. When a direct object is human or highly animate, it obligatorily receives the differential enclitic =*pe*. Crucially, =*pe* is not an Accusative suffix born *ex nihilo*; it is the language's general locative postposition, literally meaning 'in' or 'inside' (e.g., *kotý-pe* 'in the room', *Paraguay-pe* 'in Paraguay', see Estigarribia 2020: 121). To extract the salient object from the eventive core, the syntax *physically* encloses it within a spatial locative container (the Inclusion vector $\subseteq$).

(20) Guaraní Locative and DOM Syncretism

a. Spatial Locative (Zonal Inclusion $\subseteq$):

Che a-reko peteĩ óga Paraguay-pe.
1SG 1SG.ACT-have one house Paraguay-LOC
'I have a house in Paraguay.' (Velázquez-Castillo 1996: 80)

b. Differential Object Marking (DOM as $\subseteq$)

Juan o-echa aguara-pe.
Juan 3.ACT-see jaguar-DOM(LOC)
'Juan sees a jaguar.' (Irimia & Roessler 2020: 258)

Furthermore, this spatial logic extends to the prefixal domain (Agreement). Guaraní employs an active/inactive alignment, where the patients (objects) of transitive verbs are encoded via the inactive prefix paradigm. Strikingly, these inactive prefixes (e.g., *che-* '1SG.INACT') are morphologically identical to the prefixes used for nominal possession (Velázquez-Castillo 1996).

(21) Guaraní Possessive and Inactive Object Syncretism

a. Nominal Possession (Possessor as $\subseteq$)

che-róga
1SG.INACT-house 'my house' (Estigarribia 2020: 120)

b. Inactive Object Agreement (Patient as $\subseteq$)

Nde che-nupã.
2SG 1SG.INACT-beat '

You beat me.’ (Velázquez-Castillo 1996: 41)

As established by Manzini and Franco (2016), possession is the ‘core’ materialization of Zonal Inclusion ($\subseteq$) within the nominal domain. Therefore, Guarani does not compute abstract "Object Agreement"; it simply encapsulates the patient as the topological "possessor" of the event. In Guarani, what you see (a Locative adposition or a Possessive prefix) is exactly what you get. The entire category of "structural Case" is thus revealed to be a theoretical construct: the grammar manipulates caseless roots (Nominative) and spatial-topological boundaries (DOM/Obliques).

4.1.2 Palauan: The Prepositional Nature of DOM and the Defeat of Abstract Probes

Formalist approaches to Differential Object Marking frequently appeal to Austronesian languages like Palauan to argue that object marking is the result of abstract Case licensing and ϕ -feature computation, completely divorced from topological realities (e.g., Levin 2019). Palauan exhibits a strict aspectual split regarding object licensing: while perfective verbs overtly register object agreement, imperfective verbs require specific or human objects to be preceded by the marker *er*, which is treated as the realization of an abstract [Accusative] feature assigned to highly ranked arguments.

However, a closer morphological inspection dismantles again a purely structural analysis, perfectly aligning Palauan with our WYSIWYG principle. As established by Georgopoulos (1985), the differential object marker *er* is not as a dedicated Accusative particle. It is, in fact, the language's multipurpose, general preposition (P). Crucially, the exact same preposition *er* that shields salient objects from verbal incorporation in imperfective clauses is also the preposition used to mark the Possessor within the nominal domain. Georgopoulos (1985) provides a clear example where the very same morpheme performs both functions in the same clause.

(22) Palauan DOM and Nominal Possession Syncretism

ak-uleldanges	er	a	resensei	er	ngak
1SG.IM-honor	PREP(DOM)	PL	teacher	PREP(GEN)	1SG

‘I respected my teachers.’ (Georgopoulos 1985: 64)

This morphological identity between the DOM marker and the Possessive/Oblique adposition provides evidence for the spatial-relational architecture of the clause. In order to prevent the topological fusion of a specific, salient object with the unbounded eventive mass of an imperfective verb, Palauan syntax does not value an uninterpretable Case feature. Instead, it encapsulates the

object within a prepositional boundary, mapping it as the topological "possessor" or locative target of the event (the Inclusion vector $\subseteq$). What formalists describe as the valuation of an abstract Case probe is, materially and syntactically, the deployment of an overt spatial preposition to enforce the Anti-Incorporation Ban.

4.2 "Global Case Splits" and the Absence of Look-Ahead

"Global Case Splits" are frequently brandished as the ultimate proof for abstract functional calculus. In languages exhibiting global splits, the morphological marking of an argument purportedly depends on the properties of both the subject and the object. For instance, as we have already seen, in the Papuan language Awtuw, as discussed by Bárány (2018) based on Feldman (1986), the direct object receives overt differential marking (realized by the suffixes *-re/-te/-e* on common nouns) only if it outranks or equals the subject on the empathy/animacy hierarchy.

Such patterns necessitate complex algorithmic machinery: functional heads like v or T must supposedly delay Case assignment, execute multiple cyclic *Agree* operations, and employ "look-ahead" mechanisms to compare the ϕ -features of both arguments before calculating which one receives a dependent Case feature. Under our spatial-relational framework, this algorithmic operation is entirely superfluous. The global split in Awtuw is not the result of a ϕ -probe scanning the clause from a distance; it is a direct, local topological reaction to a *collision of salience*. When an internal argument is as animate or more animate than the agent, as we have seen in (4), it possesses an intrinsic referential gravity that resists conceptual absorption into the eventive core.

To prevent the object from structurally overpowering the agent and disrupting the natural origin-to-target causal vector, the syntax is forced to bound the object, deploying the spatial container of Zonal Inclusion ($\subseteq$). The deployment of DOM in Awtuw is therefore a purely local geometric necessity triggered by the imminent threat of topological fusion, requiring no abstract "look-ahead" computation.

Furthermore, the topological nature of the Awtuw DOM marker *-re* is confirmed by its behavior beyond the nominal domain. As Feldman (1986: 57) documents, the exact same suffix *-re* (and its allomorphs) is deployed to mark finite complement clauses embedded under verbs of saying and propositional attitudes, as illustrated in (23). In standard Minimalist terms, assigning an abstract [Accusative] Case feature to a fully inflected finite clause is theoretically problematic, as CPs do not require structural Case to satisfy the Case Filter.

- (23) *rey* *Numoy-re* *da-mak-e* *Liwmiy-e* *d-ae-y-e-re*
 He *Numoy-OBJ* *FA-say-PST* *Lumi-LOC* *FA-go-PST-OBJ*

‘He told Numoy that someone went to Lumi’ (Awtuw)

Under our geometric approach, however, this isomorphism is predicted. The relator $\subseteq$ is not a category-specific nominal diacritic, but a universal topological boundary. Just as highly animate DPs threaten to fuse with the verbal root and must be encapsulated by $\subseteq$ to prevent pseudo-incorporation, an entire embedded eventive space (a complement clause) *threatens* to topologically collapse into the matrix event. To maintain the two eventualities separate, the syntax recruits the exact same spatial boundary ($\subseteq$, spelled out as *-re*) to encapsulate the subordinate clause, mapping it as the topological target of the matrix predicate. What it is commonly stipulated as two entirely different phenomena, Differential Object Marking and clausal complementation, are revealed by the WYSIWYG principle to be, in various languages, the same geometric operation.⁹

Bárány (2018) also heavily relies on the global case split in Sahaptin, where 3rd person subjects receive differential marking (the 'inverse ergative' *-nim*) only when acting upon 1st or 2nd person objects, to argue for a purely functional calculus based on ϕ -probes, seemingly devoid of any spatial reality. Actually, a rigorous diachronic and morphological analysis dismantles this assumption. As demonstrated by Rude (1985: 145), the Sahaptin inverse/ergative marker is etymologically and formally identical to the verbal *Cislocative* directional affix (indicating motion 'hither', namely towards the deictic center).

The diachronic trajectory is transparent: *Verbal Cislocative* > *Nominal Cislocative (Inverse)* > *Ergative*. Geometrically, when a 3rd person acts upon a Speech Act Participant (SAP), the event is conceptualized as a spatial vector directed towards the deictic origin. This inverse alignment, where the subject is marked with *-nim* only if the object is 1st or 2nd person (3>SAP.ERG), is clearly observable in authentic Sahaptin discourse. The following example perfectly illustrates this dynamic, where the 3rd person agent ('older brother') acting on a 1st person plural inclusive patient locally triggers the cislocative/ergative boundary.

(24) Sahaptin (Cislocative/Inverse Ergative marking)

yaya-nim

aw=na

i-wanpi-sha

⁹ Clearly, this topological strategy is not a typological anomaly restricted to Papuan languages, but a universal principle of clausal embedding. In Romance languages, for instance, infinitival complements are standardly introduced by the very same spatial adpositions that lexicalize the $\subseteq$ operator (e.g., Italian *vado a dormire* 'I go to sleep', *penso di venire* 'I think of coming'). As extensively noted in the literature on restructuring and clause-union (Wurmbrand 2001; Cinque 2006), these prepositional introductory elements often do not correspond to a standard C-layer and are standardly dismissed as semantically vacuous "dummy" markers. Under our geometric framework, however, they are nothing of the sort. They are literal topological boundaries ($\subseteq$) deployed to encapsulate the embedded infinitival root, shielding it from total eventive fusion (restructuring) with the matrix predicate.

older.brother-ERG(CSL) now=1PL.INC 3SG.S-call-IMPV
 ‘Now older brother is calling us.’

Thus, the Sahaptin inverse ergative is not an abstract structural Case feature born *ex nihilo*, nor is it the result of delayed *Agree* valuations. Rather, it is the direct projection of a spatial-deictic vector, mapping the external argument as the physical origin of an incoming trajectory. This perfectly corroborates our view: the marker is spatial because the syntactic relation it establishes is fundamentally topological.

4.3 Differential Agreement or Event Possession? The Challenge of Hungarian and Amharic

Formal approaches frequently employ Differential Object Agreement (DOA) to defend the existence of abstract ϕ -probes and structural Case. According to this view (e.g., Bárány 2018), languages that mark object salience directly on the verbal head rather than (or in addition to) the nominal dependent prove that differential marking is a purely functional *Agree* operation, devoid of any spatial or adpositional reality. However, that DOA is merely the *head-marking* reflex of the exact same geometric operator: Zonal Inclusion ($\subseteq$).

4.3.1 Hungarian and Turkish: The Object as the Possessor of the Event

In the typological landscape of object marking, Hungarian is frequently cited as a textbook example of Differential Object Agreement (DOA). As extensively documented by Bárány (2018), Hungarian transitive verbs exhibit a rigid split in their agreement paradigms. When the direct object is indefinite or non-specific, the verb takes the subjective (indefinite) conjugation, registering only the subject's features. However, when the object is definite or highly referential, the verb obligatorily shifts to the objective (definite) conjugation.

(25) Hungarian (Differential Object Agreement)

a. Subjective Conjugation (Indefinite Object):

Én lát-ok egy kutyá-t.
 1SG see-1SG.SBJ a dog-ACC

‘I see a dog.’ (Bárány 2018: 2)

b. Objective Conjugation (Definite Object)

Én lát-om a kutyá-t.
 1SG see-1SG.OBJ the dog-ACC

‘I see the dog.’ (Bárány 2018: 2)

Bárány treats this objective agreement as an algebraic operation: an abstract ϕ -probe on v successfully values an uninterpretable feature (such as [DEF] or [PERSON]) located on the D head of the object. In these models, the actual morphological shape of the objective agreement suffix is treated as an arbitrary spell-out of these abstract formal features.

Yet, a closer look at the morphological substance reveals a profound structural isomorphism that defies an algebraic explanation. As originally noted by Szabolcsi (1994) and further systematized by Trommer (2005) (see Bárány 2018: 78-79, Kiss 2017), the objective agreement suffixes on the Hungarian verb are morphologically parallel, and historically related, to the possessive suffixes in the nominal domain. When a singular subject acts upon a definite object, the verbal suffix is identical to the suffix denoting a singular possessor on a noun.

(26) Hungarian (Isomorphism of Possession and DOA)

a. Nominal Possession:

ház-am

house-1SG.POSS

‘my house’ (Trommer 2005, cited in Bárány 2018: 78)

b. Objective Agreement:

lát-om

see-1SG.OBJ

‘I see it/him/her’ (Trommer 2005, cited in Bárány 2018: 78)

The fact that the syntax utilizes the exact same morphological exponent for nominal possession and transitive object agreement is an inexplicable accident. Why would a ϕ -probe checking a [PERSON] feature on a direct object spontaneously spell out as a nominal possessor?

Under our spatial-relational architecture, this is not a morphological accident at all, but the direct reflection of event geometry (Manzini & Franco 2016). In Hungarian, the universal object suffix -t is a generalized, historically opaque lative/locative marker that has lost its primary anti-incorporation function. Therefore, to shield a highly salient, definite object from undergoing pseudo-incorporation into the eventive core, the Hungarian syntax must recruit an active topological

operator. It deploys the elementary predicate of Zonal Inclusion ($\subseteq$), mapping the object as the topological "possessor" of the eventive space.¹⁰

The mechanism is identical to the one observed in Turkish, where the DOM marker *-i* is literally the 3rd person possessive suffix (Kornfilt 1997). The only typological difference is that, while Turkish spells out the possessive ($\subseteq$) vector directly on the object DP, Hungarian spells it out as an agreement dependency on the verbal head. There is no invisible ϕ -probe driving this derivation; there is only the overt manifestation of the possessive geometry. The WYSIWYG (What You See Is What You Get) principle dictates that if the marker looks like a possessor, it acts topologically as a possessor. Transitivity, in this domain, is explicitly formatted as the possession of the event.

4.3.2 Amharic: Clitic Doubling as a Topological Trace

The same topological logic can be applied to reframe the formalist arguments based on clitic doubling in Semitic languages like Amharic. In Amharic, direct objects split neatly along the lines of definiteness: highly referential (definite) objects obligatorily receive the overt DOM suffix *-n* (or *-in*). Crucially, as extensively documented by Amberber (2008) and Kramer (2014), this differential marking systematically triggers object clitic doubling on the verbal head.

The minimal contrast is clear: when the object is indefinite and unmarked, the verb exhibits solely subject agreement (27a). However, when the object is highly individuated and forced to take the DOM suffix *-n*, an object clitic obligatorily emerges on the verbal complex (27b-c).

¹⁰ Actually, the simultaneous presence of object agreement/clitics on the verb and morphological case/adpositions on the noun (as seen in Hungarian DOA + *-t* suffix, or Spanish *léismo* doubling the *a*-marked object) need not be analyzed as a redundancy where one element is reduced to an opaque morphological reflex. Drawing on Manzini et al. (2020), this "double marking" is the natural consequence of how geometric operators function. As argued for instances of *Suffixaufnahme* (case stacking), the Zonal Inclusion operator ($\subseteq$) does not merely project a single node, but defines an entire topological domain. Every substantive item merged within that domain can structurally reflect its inclusion by bearing the corresponding morphological signature. Differential Object Marking can thus be achieved through head-marking (on the verbal spine) as well as dependent-marking (on the object). In languages that exhibit this "double shield" (like Neapolitan, which can display up to three simultaneous layers of object marking), the same lexicalization conditions that require the ($\subseteq$) boundary in the predicative domain (the full DP) simultaneously drive its expression in the inflectional domain (the clitic/agreement). The *oblique* object is thus completely *sealed off* from the eventive core by a bipartite topological boundary, preventing pseudo-incorporation. Further notice that Recently, Haspelmath (2026) has argued against the typological unity of Differential Object Marking, proposing to split the phenomenon into Differential Object Flagging (dependent-marking via case/adpositions) and Differential Object Indexing (head-marking via agreement/clitics), on the premise that they obey distinct functional motivations. However, under the present spatial-relational framework, this split introduces an unnecessary theoretical fragmentation. As shown by our "Double Shield" analysis (e.g., in Amharic, Hungarian, or Iberian Spanish), flagging and indexing are not unrelated typological accidents, but symmetric morphological externalizations of the exact same geometric boundary (such as the $\subseteq$ or $\supseteq$ relators) erected to prevent pseudo-incorporation.

(27) Amharic (The DOM and Clitic Doubling Correlation)

a. Unmarked Object (No Clitic)

Almaz doro wät' bäll-at[ɬf]
 Almaz.F chicken stew eat-SBJ.3SG.F

‘Almaz ate chicken stew.’ (Kramer 2014: 601)

b. Marked Object (DOM + Clitic)

Almaz doro-w-in bäll-ät[ɬf-iw]
 Almaz.F chicken-DEF-DOM eat-SBJ.3SG.F-OBJ.3SG.M

‘Almaz ate the chicken stew.’ (Kramer 2014: 601)

c. Marked Object (DOM + Clitic)

si'ota-w Aster-in as-dəssət-at
 gift-POSS.3M Aster-DOM CAUS-happy.PRF.3M-OBJ.3SG.F

‘His gift made Aster happy.’ (Amberber 2008: 751)

It is possible to analyze this doubling as a secondary effect of abstract Case licensing and strictly functional *Agree*, effectively equating the clitic to a meaningless agreement affix checking uninterpretable ϕ -features on v . However, syntactic diagnostics falsify this algebraic assumption. As Baker and Kramer (2018) demonstrate, the Amharic object marker exhibits severe referential restrictions typical of pronominal elements, completely alien to canonical agreement. For instance, the object clitic cannot double a universally quantified DP without triggering a Weak Crossover violation, nor can it double a reflexive anaphor without violating Principle B.

(28) Amharic (The Pronominal Nature of the Clitic)

*Lämma hullu-n-imm säw ayy-ä-(w)
 Lemma.M every-DOM-FOC person see.PRF-SBJ.3SG.M-(*OBJ.3SG.M)

‘Lemma saw everyone.’ (Baker & Kramer 2018)

If the clitic were the morphological reflex of a ϕ -probe on v , it should freely cross-reference quantified nominals, just as subject agreement does. Its inability to do so proves that the clitic is a true referential pronoun, not an empty inflectional feature.

Under the WYSIWYG (*What You See Is What You Get*) principle, this dual manifestation of DOM and clitic doubling is not the mechanical checking of an uninterpretable [uCase] feature, but the physical proof of the Anti-Incorporation Ban. When an object reaches a high degree of individuation (e.g., definiteness), it geometrically resists pseudo-incorporation into the verbal root.

To protect its referential integrity, the syntax is forced to physically extract it from the eventive core, bounding it with the spatial relator $\subseteq$ (spelled out as the suffix *-n*).

Because the argument is materially displaced from its canonical internal position, the syntactic space it leaves behind must be saturated. The clitic on the verb is nothing but the literal topological trace of this displacement, a pronominal boundary left on the eventive root to explicitly saturate the argumental space and prevent the bare root from undergoing fusion. Thus, both head-marking (clitic doubling) and dependent-marking (DOM) are symmetric physical instantiations of the exact same geometric necessity: erecting a topological boundary to shield a salient participant (see fn. 10).

4.3 Aspectual DOM as Bounding Geometry

The aspectual object marking split in Uralic languages, particularly Finnish and Estonian, is sometimes taken as the ultimate refutation of spatial / semantic accounts of DOM, and as a proof for the existence of purely structural Case (e.g., Kiparsky 1998; Bárány 2018). In Finnish, the morphological case of the direct object alternates based on the telicity (boundedness) of the event: atelic events take a Partitive object, while telic events require what traditional "Accusative" case. Since this alternation is driven by aspectuality rather than the animacy or referentiality of the internal argument, formalists argue that it must be the result of a purely syntactic feature valuation (e.g., an uninterpretable [uCase: Acc] feature assigned by a [+telic] *v* or Asp functional head).

Applying the WYSIWYG (What You See Is What You Get) principle to the morphological substance dismantles this assumption. In modern Finnish, the so-called singular "Accusative" does not have a dedicated morphological exponent; it is systematically syncretic with the Genitive suffix *-n* (Karlsson 2018). Consider the classic aspectual minimal pair:

(29) Finnish (Aspectual Split as Bounding Geometry)

a. Atelic/Unbounded event (Partitive = Separation/Incompleteness):

Mies luki kirja-a.
man.NOM read.PST.3SG book-PART

‘The man was reading a/the book.’

b. Telic/Bounded event (Genitive as Zonal Inclusion $\subseteq$):

Mies luki kirja-n.
man.NOM read.PST.3SG book-GEN

‘The man read the book (completely).’

Under our spatial-relational framework, this is not an accidental morphological syncretism, nor is it the spell-out of an abstract structural Case. The Genitive is the direct nominal externalization of Zonal Inclusion ($\subseteq$). When an event is atelic (29a), it is topologically unbounded and open-ended; therefore, the syntax cannot map the object as the bounding limit of the event. Instead, the object is marked with the Partitive, a case that historically derives from an ablative/separative spatial relator, denoting partial or ongoing (unbounded) topological involvement.

Conversely, when an event is telic (29b), it is conceptualized as a fully bounded, delimited topological space. To anchor the object to this bounded eventive structure, the syntax physically maps it as its "possessor", encapsulating it with the $\subseteq$ relator (spelled out as the Genitive *-n*). What generative grammar describes as a structural "Accusative" triggered by aspect, is physically and geometrically just the Possessor of a completed event. The Uralic aspectual split is thus not a counterexample to the spatial nature of object marking; it is its most spectacular confirmation. By taking the event's bounding geometry into account, we eliminate the need for aspectual functional probes and completely absorbs structural Case into the domain of topology.¹¹

4.4 Focus on Person Splits: Deictic Centers in Kashmiri and Chukchi

4.4.1 Kashmiri: From Abstract Probes to Interpretable Topologies

Kashmiri (Indo-Aryan) is presented as a proof for the necessity of multiplying abstract functional probes to calculate Case and Agreement. Based on the descriptive grammar of Wali and Koul (1997), Bárány (2018) proposes a highly sophisticated architecture for Kashmiri involving three distinct ϕ -probes (on T, v, and Appl). This 'cartographic' proliferation is invoked to explain a complex intersection of splits: Kashmiri subjects receive Ergative marking only in the perfective aspect, while direct objects receive Dative marking (DOM) in the imperfective aspect based on a $1 > 2 > 3$ person hierarchy. Furthermore, Kashmiri verbs display a rich paradigm of pronominal suffixes cross-referencing these arguments.

Under our spatial-relational view, the Kashmiri system requires no abstract probes, no derivation-crashing uninterpretable $[u\phi]$ features, and no Impoverishment rules. Its morphosyntax is the direct manifestation of spatial vectors tracking event boundaries and deictic centers.

¹¹ Kiparsky (1998) argue that the Finnish Partitive/Accusative alternation is treated as the assignment of abstract structural features driven by aspectual heads. We argue that this split is purely topological. The Finnish "Accusative" spells out the Zonal Inclusion operator ($\subseteq$), mapping the object as the bounded limit or "possessor" of a completed telic event. Conversely, the Partitive, which historically derives from an Ablative/Separative marker, possibly spells out the Reverse Inclusion vector ($\supseteq$). It is deployed precisely when the event fails to reach a definitive boundary (in atelic or negative contexts), framing the object not as a reached goal, but as a spatial origin or mass from which the action abstracts a part. Thus, the Uralic case split is not a functional alternation, but the material manifestation of the transitive vector either reaching its boundary ($\subseteq$) or failing to do so ($\supseteq$).

First, the aspectual split is purely geometric. The Ergative case is strictly confined to the perfective aspect. As shown in (30), perfectivity entails a bounded spatial trajectory (examples are taken from Wali and Koul 1997).¹²

(30) Kashmiri (Perfective: Ergative Subject)

me por mazmu:n
 1.ERG read.PRF essay.ABS
 ‘I read the essay.’

To trace this telic, *ballistic* vector, the syntax requires the morphological realization of the Origin/Source vector ($\supseteq$), which surfaces as the Ergative *me*. In the imperfective, the event is an unbounded topological container, lacking a designated spatial origin; thus, the Ergative ($\supseteq$) does not emerge. No probe "switches off" or "shifts" to T; the geometry of the event simply changes.

Second, the person-driven DOM split in the imperfective is a direct reflection of the Geometry of Salience. In direct configurations, where the Agent outranks the Patient (e.g., 1st person acting on 2nd person), the direct object is unmarked (Nominative/Absolutive):

(31) Kashmiri (Imperfective Direct: Unmarked Object)

bi chu-s-ath tsi parina:va:n
 1SG.NOM be.M.SG-1SG.SBJ-2SG.OBJ2SG.NOM teaching
 ‘I am teaching you.’

In inverse configurations, however, the action moves from a peripheral entity (e.g., a 3rd person) towards the conversational core (the 1st or 2nd person, acting as the deictic center of the discourse). As shown in (32), the highly salient deictic center cannot be incorporated into the eventive mass of the verb.

¹² This morphological reality reveals that the Kashmiri perfective transitive is conceptually and geometrically similar to the "ethical dative" or "dative of interest" constructions found in Romance. When an Italian speaker says *Io mi leggo il libro* (lit. 'I read to-me the book'), the dative clitic *mi* is not assigned by a structural Case probe; rather, it is an interpretable spatial operator that anchors the *reading* event within the personal sphere (the inclusion zone) of the salient participant. The Kashmiri sentence in (30) does the same thing. Because the perfective aspect bounds the event into a completed, detached topological whole, the syntax cannot map the Agent directly onto the eventive spine. Instead, it must tether the Agent to the completed event using a relator. The pronoun *me* in Kashmiri simply spells out this spatial tethering, formally stating: "With respect to me / in my topological sphere, there is the reading of the essay". By treating the Kashmiri *me* as a structural feature rather than an interpretable spatial anchor, traditional alignment typologies obscure the geometric similarity between Indo-Aryan split-ergativity and Romance dative constructions.

(32) Kashmiri (Imperfective Inverse: Dative DOM)

su	chu-y	tse	parina:va:n
3SG.NOM	be.M.SG-2SG.OBJ	2SG.DAT	teaching
‘He is teaching you.’,			

To prevent this fusion, Kashmiri syntax shields the object with the spatial vector of Inclusion/Terminus ($\sqsubseteq$), overtly lexicalized as the Dative case. Finally, the complex system of verbal suffixes in Kashmiri (such as *-s* and *-ath* in (31), or *-y* in (32)) does not represent the valuation and deletion of uninterpretable features by three probes. Crucially, as established by Manzini and Savoia (2007), agreement markers and pronominal clitics are not meaningless formal diacritics; rather, they are fully interpretable heads endowed with intrinsic denotational and relational properties. In a V2 language like Kashmiri, the extracted arguments, shielded by their respective topological boundaries ($\supseteq$ / $\sqsubseteq$), must be tracked within the eventive core. The verbal suffixes are the lexicalization of these extracted referents, acting as interpretable clitics that satisfy the argumental slots of the predicate.

Agreement in Kashmiri, therefore, is not an operation of feature-deletion, but a relation of identity holding between two sets of valued, interpretable features (the clitic on the verb and the dislocated spatialized DP).¹³

4.4.2 Inverse Markers as Spatial Vectors in Chukchi

In formal generative typology, the Paleosiberian language Chukchi is frequently invoked to justify the existence of complex post-syntactic operations, particularly Impoverishment rules manipulating abstract ϕ -features. Bárány (2018), building on Bobaljik & Branigan (2006), focuses on the so-called "spurious antipassive" (SAP). In Chukchi inverse configurations (e.g., when a 3rd person subject acts upon a 1st or 2nd person object), the verb obligatorily takes the affixes *ine-* or *-tku*. Although these affixes generally mark the antipassive derivation, in the SAP the clause remains

¹³ A similar critique applies to contemporary accounts of Eastern Neo-Aramaic (e.g., Kalin & van Urk 2015, Kalin 2018), where complex aspectual alignment splits are modeled by proliferating abstract functional machinery, such as postulating multiple ϕ -probes or stipulating complex "partial agreement reversal" algorithms. However, these purely algebraic solutions completely ignore the diachronic and synchronic morphological substance of the language. As extensively documented by Coghill (2016), the Neo-Aramaic past perfective system is built on the historical *qtil-li* construction. Because Aramaic lacks a transitive verb 'to have', possession is spatially expressed via oblique predication utilizing the dative/oblique preposition *l-*. The "ergative" agent in these past tenses is obligatorily marked by the so-called "L-suffixes", which are nothing but the direct morphophonological continuation of this dative/possessive preposition. Under the WYSIWYG principle, the Neo-Aramaic ergative is not an invisible structural Case valued by a ϕ -probe; it is literally the spatial possessor of the event. The syntax maps the Agent as the topological possessor (the $\sqsubseteq$ vector) of the resultative eventive mass. What Kalin & van Urk (2015) model as functional "agreement reversal" is simply the geometric reflex of a possessive predication. Any attempt to superimpose an abstract structural Case assignment theory on this transparent spatial relation completely misses the empirical reality of the language.

morphosyntactically and semantically transitive. To account for this, Bárány (2018) stipulates an abstract Impoverishment rule that deletes the ϕ -features on the *v* head, tricking the morphology into inserting intransitive/antipassive exponents. This mechanism is illustrated by the contrast between a direct configuration in (33a) and an inverse one in (33b) (taken from Bobaljik & Branigan 2006).

(33) Chukchi Spurious Antipassive (SAP)

a. *Direct (1SG > 2SG): Regular transitive agreement*

γəm-nan	γət	tə-lʔu-γət.
1SG-ERG	2SG.ABS	1SG.SUBJ-see-AOR.2SG.OBJ

'I saw you.'

b. *Inverse (3SG > 1SG): SAP with 'ine-' prefix*

ə-nan	γəm	ine-lʔu-γʔi.
3SG-ERG	1SG.ABS	AP-see-AOR.3SG

'S/he saw me.'

However, under the spatial-relational architecture, this stipulative machinery is revealed to be a theoretical artifact. As Polinsky (2017) observes, "the Chukchi 'spurious' antipassive is not antipassive at all" and has nothing to do with valency-changing morphology. Descriptive grammars (Dunn 1999: 182) *ine-* and *-tku* in these contexts as overt markers of inverse alignment. Geometrically, when a peripheral 3rd person acts upon a Speech Act Participant (the deictic center), the event traces an incoming spatial trajectory. The marker *ine-* is not the result of deleted uninterpretable features, but rather the fully interpretable materialization of this inverse spatial vector, deployed to shield the highly salient deictic core from eventive fusion.

Furthermore, Chukchi syntax provides striking independent evidence against the view that Ergative is a core structural Case assigned by T. As extensively documented by Polinsky (1994, 2017), Chukchi exhibits an extraction asymmetry: while Absolutive arguments (both transitive patients and intransitive subjects) can be straightforwardly relativized with a gap, the direct extraction of an Ergative agent is strictly ungrammatical. To relativize the agent, the clause must undergo full antipassivization, stripping the agent of its Ergative marker and demoting the patient. In our geometric framework, this restriction is perfectly predictable. The extraction asymmetry is robustly attested in the paradigm below. While the absolutive object extracts (34b), attempting to relativize the ergative subject with a gap results in ungrammaticality (34c) (examples taken from Polinsky 2017).

(34) Chukchi Extraction Asymmetry

Tumg-e	ŋinqey	rəyegtetew-nin.
friend-ERG	boy.ABS	save-AOR.3SG.3SG

b. *Relativization of ABS object (Grammatical)*

... [<i>t_i</i> rəyagtala-lʔ-ən]	ŋinqey _{<i>i</i>}
save-PTCP-ABS	boy.ABS

c. Relativization of ERG agent

*[<i>ti</i> ɲɪnqey	rəyagtala-lʔ-ən]	tumgətumi
boy.ABS	save-PTCP-ABS	friend.ABS (Intended: 'the friend that saved the boy')

5. Conclusion

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For decades, formal linguistics has treated structural Case as the invisible, uninterpretable scaffolding holding syntax together. The discipline has erected an autonomous, highly complex algebraic system of licensing algorithms, cyclic expansions, and phi-agreement probes that have become detached from conceptual, morphological, and spatial reality. In this paper, we have tried to demonstrate that this abstract formal machinery is a theoretical artifact.

The pervasive typological realities examined here, ranging from the radical morphological bareness of absolutive objects in Australian Pama-Nyungan languages to the striking locative isomorphism of Guaraní DOM, all inevitably converge on a single conclusion: Structural Case does not exist. Syntax operates, at its deepest cognitive and computational level, through the medium of spatial geometry/topology. When an internal argument is salient or highly specified enough to naturally resist absorption into the core verbal event, the grammar simply does not compute empty formal features, nor does it arbitrarily trigger post-syntactic Impoverishment rules to salvage an algebraic derivation. Instead, it systematically erects overt material boundaries to enforce the Anti-Incorporation Ban. The computational system actively deploys physical vectors of Origin ($\ni$) to trace causal sources and actors, and vectors of Zonal Inclusion ($\subseteq$) to define topological containers, experiencers, and spatial targets.

By recognizing and applying a *What You See Is What You Get* principle, we are able to unify complex and seemingly disparate morphosyntactic variation, spanning from problematic aspectual alignment splits to the intricacies of person-driven differential object marking, without resorting to stipulative parameters, silent functional heads, delayed cyclic probe expansions, or the unnecessary functional stratification found in frameworks like Nanosyntax.

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